GTPA v2 — Block C (theorem half): the v1.2 theorems, proved, degraded or bounded

Status:

```
document   = GTPA v2 / Block C theorem answers / 2026-09-26 /
             read once, findings applied
folder     = gtpa-v2/ (text of the next GTPA edition + its crosswalk from v1.2)
base       = UFD nucleus v0.1, ufd/UFD-NUCLEO.md at research
             7db1e5ef1e1e04daafab7b62538068afafd5dbf5 (blob 4f52a62c):
             convention C1-C6, statements U-01..U-15, Block A answers;
             Block B, gtpa-v2/BLOCK-B-ANSWERS.md at the same commit
             (blob 3d8dea16): statements B-01..B-11
scope      = Block C of the approved architecture, theorem half: EG-03.1,
             EG-03.2, EG-03.3, EG-03.4, EG-03.5, EG-03.6, EG-03.8, EG-03.9,
             EG-03.10, EG-03.11, and the Thm 9.1 half of EG-03.0 (its Thm 1.1
             half is answered in UFD §3.3); statements CT-01..CT-15
ceiling    = FORMAL_LOCAL_CONDITIONAL / NONPROBATIVE
claims     = NONE. Claim authority = gtpa-lab/CLAIM-LEDGER.md; nothing here
             is a GTPA claim until it passes the ledger gate (next free ID
             C0138; C0137 is held for EG-05c)
physics    = CLOSED     MATH<->GTPA = UNBOUND
producer   = Claude (Anthropic), workflow writing agent
evaluator  = one independent read done (same provider, L-A1), 2026-09-26.
             It found 2 real errors, both fixed, no verdict changed:
             (1) CT-01 / §3.1 said the holding cost depends only on G∇J;
             false when A ≠ 0 (the holding control is (G − A)∇J, and A
             adds ½(A∇J, G⁻¹A∇J) to the declared cost); corrected;
             (2) §3.6 / §3.8 said v1.2 gives no scalar coherence; §9.4
             (pp.53-54) uses Mean(C_ij), not tied to Thm 9.1 or 9.3;
             corrected. And 4 minor points, applied: CT-12's open sets
             are subsets of the basins (full basins OPEN, §5); CT-12 (e)
             restricted to other cost terms bounded below or at most
             linear in u, with its one-line proof; unique C_opt needs
             {P > 0} to be an interval; §3.2 names its reading and notes
             the O_anchor(territory) reading. A later independent read of
             gtpa-v2/GTPA-v2.md (2026-09-26) found that the plain line of
             §3.4 stated the L¹-with-cap condition as necessary; corrected
             to sufficient
checks     = no human check; no cross-provider check; no formal (Lean) check
recompute  = decisive calculations recomputed by machine on 2026-09-26: 29 agree, 3 not
mechanizable;
             the remark after CT-07 (2026-10-05) recomputed once, agrees
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Opening record. The 29 (+2) questions on GTPA v1.2 were opened by the operator on 2026-09-25 ("3, sí por supuesto"; architecture "Go"), in three blocks. Block A is answered in the UFD nucleus (UFD §3), Block B in `BLOCK-B-ANSWERS.md`. Block C holds 18 GTPA-internal questions. This file answers the ten of them that ask to prove, degrade or bound a v1.2 theorem, plus the Thm 9.1 half of EG-03.0, which the architecture moved to Block C. The other eight Block C questions (EG-02.1, EG-02.6, EG-04.2, EG-05.b, EG-07.b-src, EG-08.a, EG-10.a3, EG-10.c2) are not answered here. Working style (operator, 2026-09-26): coherent text, one status per statement, one independent read; no tests, check modules or hash pins.

0. How to read this file

- Same format as the UFD nucleus (UFD §0) and Block B (Block B §0). Every statement has an ID `CT-nn`, its types and hypotheses, one status — `DECLARED`, `PROVED_HERE` (complete proof here,

checkable by hand), `STANDARD_FACT` (exact locator) or `OPEN` — and one line in plain Spanish (`En llano`).

- Section 3 answers the eleven questions. Each has the exact question text of the triage record (research `d0cf83e`; for EG-03.6 the reviewer's corrected question, adopted by the adjudication, is quoted as well), the GTPA v1.2 locators, the theorem as printed with its types and hypotheses, exactly one verdict — `HOLDS_UNDER_STATED_HYPOTHESES` / `FAILS_WITH_COUNTEREXAMPLE` / `ILL_POSED_AS_WRITTEN` (+ the repair GTPA v2 adopts) / `OPEN` — the argument, the repair, and one sentence in plain Spanish. `CROSSWALK-v1.2.md` will carry one row per answer (the lead integrates it).
- Declared models. Most of these theorems belong to GTPA's agent layer, and v1.2 gives them no model: no update law (Thm 10.1), no formula for the cost (Thm 7.2, Thm 9.1), no collective motion (Thm 9.2), no definition of the predicates (Thm 10.2, Thm 11.1). Where an answer needs a model, it is declared inside the statement that uses it, marked `DECLARED`, and kept as small as possible. A counterexample built on a declared model refutes a universal claim that covers that model; a positive result on a declared model holds for that model only.
- The base motion is UFD's: the first-order Itô SDE (M) on R^n and, under the pairing (P), its paired variant (M_P) (C1–C5; U-01, U-02, U-08). It matters for EG-03.6 (Thm 3.1, Thm 4.1), EG-03.9 (the first-order base has no phase in which the state keeps moving without control) and EG-03.10 (curl of the base drift). UFD's notation (UFD §0) is used unchanged; §1 adds what Block C needs.
- Words: UFD's rule applies. Words from physics appear only inside quotations or paraphrases of v1.2 and in explicit negations. v1.2's own names (S_{total}, D_{met}, $E_{\text{Hamiltonian}}$, $V_{\text{effective}}$, Z, u_{opt}, `Force_Social`, C_{ij}, "Class A", "Class B", "Pulse-Coast-Brake", ...) are used as names of v1.2 objects and nothing more.
- Page numbers are PDF page indices of GTPA v1.2 (`quasar:modulos/formacion/fuentes/GTPA-v1.2.pdf`, SHA-256 $0c9434c1...865f$), the numbering of its text export (blob 32787369) and of the triage. Quotations were checked against the text export.
- No MATH object is used as proof anywhere in this file. The claim ledger was read (search for the theorem numbers and the key names): no ledger claim addresses any v1.2 theorem answered here; the one related record is noted in §3.1.

1. Notation added for Block C

- For a symmetric positive definite matrix Q: $\langle u, v\rangle_Q := u^\top Q v$ and $|u|_Q := \langle u, u\rangle_Q^{1/2}$; $\lambda_{\min}(Q)$ is its smallest eigenvalue. For a symmetric $Q \succeq 0$, $Q^{1/2}$ is its symmetric positive semidefinite square root [Horn & Johnson 2013, Thm 7.2.6].
- For a symmetric $Z \succeq 0$ with orthonormal eigenvectors w_i and eigenvalues σ_i ($Z = \sum_i \sigma_i\, w_i\, w_i^\top$), $Z^+ := \sum_{\sigma_i > 0} \sigma_i^{-1}\, w_i\, w_i^\top$ (its Moore–Penrose pseudoinverse). For such Z the singular values are the σ_i, and $Z = W\, S\, W^\top$ with $W = [w_1 \dots w_k]$, $S = \operatorname{diag}(\sigma_i)$ is a singular value decomposition.
- For an open $\Omega \subset R^n$ and $F \in C^1(\Omega; R^n)$, $\operatorname{curl} F$ is the antisymmetric $n \times n$ matrix field with entries $(\operatorname{curl} F)_{ij} := \partial_i F_j - \partial_j F_i$ (for $n = 3$ its three independent entries are the usual curl; for $n = 2$ it is the single number $\partial_1 F_2 - \partial_2 F_1$).
- The model state of v1.2 (Ax 5.1, p.33) is $y = (q_{\text{est}}, p_{\text{est}})$; for a territory state (q, p) held fixed, $y^* := (\Pi(q), J_\Pi(q)\, p)$ (v1.2's `J_Pi`, the Jacobian of Π).
- v1.2's thresholds and regularizers called "epsilon" are renamed: τ is Thm 7.1's filter threshold (ε is UFD's noise level, C4).

- argmax P is the set of maximizers of P (possibly empty).

2. Statements

CT-01 — Holding a point against the base drift [PROVED_HERE] (EG-03.6, Thm 3.1)

Setting: the noiseless base drift of (M) with an added control $u \in R^n$ (U-01 with $\Sigma \equiv 0$):

$$\dot{q} = (A - G)\nabla J(q) + u.$$

For $\bar{q} \in R^n$, the constant path $q \equiv \bar{q}$ solves this equation if and only if $u = u_{\text{hold}}(\bar{q}) \coloneqq (G - A)\nabla J(\bar{q})$ (direct: $\dot{q} = 0$).

(a) If $G(\bar{q}) \succ 0$, then $u_{\text{hold}}(\bar{q}) = 0$ if and only if $\nabla J(\bar{q}) = 0$. So for every cost c with $c(0) = 0 < c(u)$ for $u \neq 0$, holding $\bar{q}$ costs $c(u_{\text{hold}}(\bar{q})) > 0$ exactly at the points that are not critical points of J. The holding control is $(G - A)\nabla J(\bar{q})$, circulating part included; only its component along ∇J is free of A: $\langle \nabla J, u_{\text{hold}} \rangle = \langle \nabla J, G\nabla J \rangle$ whatever A.

(b) Declared cost $c(u) \coloneqq \tfrac{1}{2}\langle u, G^{-1}u \rangle$, with G constant and $\succ 0$: $c(u_{\text{hold}}(\bar{q})) = \tfrac{1}{2}\langle \nabla J, G\nabla J \rangle(\bar{q}) + \tfrac{1}{2}\langle A\nabla J, G^{-1}A\nabla J \rangle(\bar{q})$. So the circulating part adds to the cost of holding, and with $A = 0$ the cost is $\tfrac{1}{2}\langle \nabla J, G\nabla J \rangle(\bar{q})$, half the decrease rate of J along the noiseless motion at $\bar{q}$ (U-04). With $A = 0$ and $J(q) = \tfrac{1}{2}\langle q - q^*, H(q - q^*) \rangle$, $H \succ 0$, this equals $\tfrac{1}{2}\langle \bar{q} - q^*, HGH(\bar{q} - q^*) \rangle$, and it equals $k|\bar{q} - q^*|^2$ for every $\bar{q}$ (some $k > 0$) if and only if $HGH = 2kI$.

(c) Counterexample to "the holding cost is $k_{\text{thermo}}\,||\varepsilon||^2$", with ε as in v1.2's Def 3.2 and X_H typed on the UFD base as the circulating part $A\nabla J$ (U-03; v1.2 never writes X_H in terms of ∇J, UFD §3.3): $n = 1$ (so $A = 0$, the only 1×1 antisymmetric matrix), $J(x) = x^2/2$, $G = 1$, $\bar{q} = 1$. Then $u_{\text{hold}} = 1 \neq 0$, while $X_H \equiv 0$, so $\varepsilon = |X_H(q_{\text{model}}) - \text{Projection}(X_H(q_{\text{base}}))| = 0$ for every linear `Projection` and every pair of states. Every cost that is positive off 0 gives $c(u_{\text{hold}}) > 0 = k_{\text{thermo}} \cdot \varepsilon^2$, whatever k_{thermo}. The same holds in any dimension with $A \equiv 0$. With a nonzero circulating part, in $n = 2$: $A = S$ (the rotation matrix of U-03 (b); $X_H = Sq$ is the flow of $J = |q|^2/2$ for the standard symplectic form, as in UFD §3.3), $G = I$, `Projection` the identity. Then $\varepsilon = |S(q_{\text{model}} - q_{\text{base}})| = |q_{\text{model}} - q_{\text{base}}|$, while $u_{\text{hold}}(\bar{q}) = (I - S)\bar{q}$; at $q_{\text{model}} = q_{\text{base}} = \bar{q} \neq 0$, $\varepsilon = 0$ and holding $\bar{q}$ has positive cost. So ε measures the distance between the two states, not the cost of holding either of them against the slide.

Proof. (a) By U-03 (a), $\langle \nabla J, u_{\text{hold}} \rangle = \langle \nabla J, G\nabla J \rangle - \langle \nabla J, A\nabla J \rangle = \langle \nabla J, G\nabla J \rangle$, which is > 0 when $\nabla J(\bar{q}) \neq 0$ and $G(\bar{q}) \succ 0$; if $\nabla J(\bar{q}) = 0$ then $u_{\text{hold}} = 0$. (b) Write $g \coloneqq \nabla J(\bar{q})$ and substitute $u_{\text{hold}} = (G - A)g$: $\tfrac{1}{2}\langle (G-A)g, G^{-1}(G-A)g \rangle = \tfrac{1}{2}\langle g, Gg \rangle - \tfrac{1}{2}\langle g, Ag \rangle - \tfrac{1}{2}\langle Ag, g \rangle + \tfrac{1}{2}\langle Ag, G^{-1}Ag \rangle$ (G symmetric), and $\langle g, Ag \rangle = \langle Ag, g \rangle = 0$ because A is antisymmetric; with $A = 0$ only $\tfrac{1}{2}\langle g, Gg \rangle$ remains. For the quadratic J, $\nabla J(\bar{q}) = H(\bar{q} - q^*)$. The last claim is U-10 (a) (polarization) applied to the symmetric matrix $HGH/2$. (c) Direct. ∎

En llano: mantener fijo un estado cuesta exactamente donde la pendiente no es cero (hay que contrarrestar el deslizamiento y también la parte que gira); por eso la "disonancia" de GTPA, hecha solo con la parte que gira, puede valer cero mientras mantener el estado cuesta.

CT-02 — A strictly convex landscape has exactly one attractor [PROVED_HERE + STANDARD_FACT] (EG-03.1, Thm 5.1)

Let $U \subset R^k$ be nonempty, open and convex, and let $V \in C^2(U)$ be strictly convex, with compact sublevel sets $S_c \coloneqq \{y \in U : V(y) \leq c\}$ for every $c \in R$.

(a) V has exactly one critical point y_{min}; it is the unique global minimizer of V on U.

(b) Let P be a constant symmetric positive definite $k \times k$ matrix. Every solution of $\dot{y} = -P\nabla V(y)$ with $y(0) \in U$ exists for all $t \geq 0$, stays in U and converges to $y_{\min}$.

(c) Let M be constant, symmetric and positive definite, and Γ constant with $\Gamma + \Gamma^{\mathsf{T}} \succ 0$. Every solution of $M\ddot{y} = -\nabla V(y) - \Gamma\dot{y}$ with $y(0) \in U$ exists for all $t \geq 0$, stays in U, and $(y, \dot{y}) \to (y_{\min}, 0)$.

(d) The form of Thm 5.1. Let $V(y) \coloneqq \tfrac{1}{2}\langle y - y^*, K(y - y^*)\rangle - \mu \ln O(y)$ with K symmetric positive definite, $y^* \in R^k$, $\mu > 0$, $O \in C^2(U)$, $O > 0$ on U, $-\ln O$ convex on U (for instance O concave), and $O(y) \to 0$ as $y \to \bar{y}$ within U, for every $\bar{y} \in \partial U$ (void if $U = R^k$). Then V satisfies the hypotheses above. For $E_{\text{Hamiltonian}}$ read on the model state $y = (q_{\text{est}}, p_{\text{est}})$ with the territory data fixed and G_{model}, M_{cog} constant (§3.2), $E_{\text{Hamiltonian}}(y) = \tfrac{1}{2}\langle y - y^*, K(y - y^*)\rangle$ with $K = \operatorname{diag}(G_{\text{model}}, M_{\text{cog}}^{-1})$, which is positive definite if and only if $G_{\text{model}} \succ 0$ and $M_{\text{cog}} \succ 0$.

(e) Without derivatives. If U is bounded and convex, O is continuous on the closure of U, positive on U and zero on ∂U, and $-\ln O$ is convex on U, then V of (d) has exactly one minimizer on U. This covers $O(y) \coloneqq$ the distance from y to $R^k \setminus U$ — v1.2's "distance to a cliff" (Def 5.2, p.34) — which is concave on a convex U.

Proofs. (a) For $y_0 \in U$, the set $S_{V(y_0)}$ is nonempty and compact, so V attains its minimum over U at some $y_{\min}$ in it; U is open, so $y_{\min}$ is a critical point. For a differentiable convex V, $V(z) \geq V(y) + \langle \nabla V(y), z - y\rangle$, so every critical point is a global minimizer, and strict convexity allows only one minimizer. ∎

(b) $-P\nabla V$ is C^1, hence locally Lipschitz, so there is a unique maximal solution. Along it $d/dt\, V(y_t) = -\langle \nabla V, P\nabla V\rangle \leq 0$, so y_t stays in the compact set $S_{V(y_0)} \subset U$, and the solution exists for all $t \geq 0$ [Khalil 2002, Thm 3.3]. By LaSalle's invariance principle [Khalil 2002, Thm 4.4], applied on the compact positively invariant set $S_{V(y_0)}$, y_t approaches the largest invariant subset of $\{\langle \nabla V, P\nabla V\rangle = 0\} = \{\nabla V = 0\} = \{y_{\min}\}$. ∎

(c) With $w \coloneqq \dot{y}$, the function $\Psi(y, w) \coloneqq \tfrac{1}{2}\langle w, Mw\rangle + V(y)$ satisfies $d\Psi/dt = \langle w, -\nabla V - \Gamma w\rangle + \langle \nabla V, w\rangle = -\tfrac{1}{2}\langle w, (\Gamma + \Gamma^{\mathsf{T}})w\rangle \leq 0$. The sets $\{\Psi \leq h\}$ are compact subsets of $U \times R^k$: on them $y \in S_h$ and $\tfrac{1}{2}\langle w, Mw\rangle \leq h - V(y_{\min})$. As in (b), solutions exist for all $t \geq 0$ and approach the largest invariant subset of $\{w = 0\}$; on an invariant subset $w \equiv 0$, so $\dot{w} \equiv 0$ and $\nabla V(y) = 0$, i.e. $y = y_{\min}$. ∎

(d) The quadratic term has Hessian $K \succ 0$ and $-\mu \ln O$ is convex, so V is strictly convex. If O is concave and positive, $-\ln O$ is convex: for $y, z \in U$ and $\theta \in [0, 1]$, $-\ln O(\theta y + (1-\theta)z) \leq -\ln(\theta O(y) + (1-\theta)O(z)) \leq \theta(-\ln O(y)) + (1-\theta)(-\ln O(z))$, because $-\ln$ is decreasing and convex. Sublevel sets are bounded: fix $y_0 \in U$; convexity gives $-\ln O(y) \geq -\ln O(y_0) + \langle g, y - y_0\rangle$ on U, with $g \coloneqq \nabla(-\ln O)(y_0)$, so $V(y) \geq \tfrac{1}{2}\lambda_{\min}(K)|y - y^*|^2 + \mu(-\ln O(y_0) + \langle g, y - y_0\rangle) \to \infty$ as $|y| \to \infty$. Sublevel sets are closed in R^k: if $y_j \in S_c$ and $y_j \to \bar{y}$, then $\bar{y} \notin \partial U$ (there $O(y_j) \to 0$, so $-\mu \ln O(y_j) \to \infty$ while the quadratic term stays bounded), so $\bar{y} \in U$ and $V(\bar{y}) \leq c$ by continuity. The formula for $E_{\text{Hamiltonian}}$: with (q, p) fixed, $\varepsilon_q = \Pi(q) - q_{\text{est}}$ and $\varepsilon_p = J_\Pi(q)p - p_{\text{est}}$ (Ax 5.1) are $y^* - y$ split in two blocks; the inverse of a positive definite matrix is positive definite. ∎

(e) V is continuous on U, tends to $+\infty$ at ∂U and U is bounded, so its sublevel sets are compact and it attains its minimum; it is strictly convex, so the minimizer is unique. The distance $\delta(y) \coloneqq \operatorname{dist}(y, R^k \setminus U)$ is concave on a convex U: if the open balls $B(y, r)$ and $B(z, s)$ lie in U, then so does $B(\theta y + (1-\theta)z, \theta r + (1-\theta)s)$ (write a point of it as $\theta(y + w_1) + (1-\theta)(z + w_2)$ with $|w_1| < r$, $|w_2| < s$, and use convexity), so $\delta(\theta y + (1-\theta)z) \geq \theta\delta(y) + (1-\theta)\delta(z)$. ∎

En llano: si el paisaje es estrictamente convexo y sube sin límite hacia el borde de la zona segura y hacia el infinito, tiene un solo punto de reposo, y todas las trayectorias que bajan por él acaban ahí; con el error cuadrático de GTPA eso ocurre, por ejemplo, cuando el margen de seguridad tiene forma de cúpula, como la distancia al borde.

CT-03 — A log barrier over a convex safe zone with two minimizers [PROVED_HERE + STANDARD_FACT] (EG-03.1, Cor 5.1)

$d = 1$, $y = (q_{\text{est}}, p_{\text{est}}) \in R^2$, $G_{\text{model}} = 1$, $M_{\text{cog}} = 1$, territory data with $\Pi(q) = 0$ and $J_\Pi(q)p = 0$ (so $y^* = 0$ and $E_{\text{Hamiltonian}}(y) = \tfrac{1}{2}q_{\text{est}}^2 + \tfrac{1}{2}p_{\text{est}}^2$), and the Regime B barrier $-\mu \ln O$ with $\mu = 1$ and

$$O(y) \coloneqq (1 - q_{\text{est}}^2) \cdot \exp((5/2)\, q_{\text{est}}^2).$$

The safe zone $\{O \geq 0\} = [-1, 1] \times R$ is convex, $O > 0$ on $U \coloneqq (-1, 1) \times R$, and $O \to 0$ at ∂U. Then $V_{\text{effective}} = E_{\text{Hamiltonian}} - \ln O$ is

$$V(y) = -2\, q_{\text{est}}^2 - \ln(1 - q_{\text{est}}^2) + \tfrac{1}{2}\, p_{\text{est}}^2 \qquad \text{on } U,$$

and it has exactly three critical points: the two strict global minimizers $(\pm 1/\sqrt{2}, 0)$, where $V = \ln 2 - 1$, and the saddle $(0, 0)$, where $V = 0$. Each minimizer is an asymptotically stable equilibrium of $\dot{y} = -\nabla V(y)$: there are two attractors, not one.

Proof. $\partial V/\partial p_{\text{est}} = p_{\text{est}}$, and $\partial V/\partial q_{\text{est}} = -4q + 2q/(1 - q^2) = 2q(1/(1 - q^2) - 2)$ (writing $q = q_{\text{est}}$), which vanishes iff $q = 0$ or $q^2 = \tfrac{1}{2}$. $\partial^2 V/\partial q_{\text{est}}^2 = -4 + 2(1 + q^2)/(1 - q^2)^2$ equals 8 at $q^2 = \tfrac{1}{2}$ and -2 at $q = 0$; $\partial^2 V/\partial p_{\text{est}}^2 = 1$; the mixed derivative is 0. $V(\pm 1/\sqrt{2}, 0) = -1 - \ln \tfrac{1}{2} = \ln 2 - 1$. Since $V \to +\infty$ as $|q_{\text{est}}| \to 1$ or $|p_{\text{est}}| \to \infty$, the minimum of V is attained at a critical point, and the two minimizers have the same value. At a critical point y_m with positive definite Hessian, y_m is an isolated critical point, $V - V(y_m) > 0$ and $d/dt\,(V - V(y_m)) = -|\nabla V|^2 < 0$ on a punctured neighbourhood of y_m, which gives asymptotic stability [STANDARD_FACT: Lyapunov's theorem, Khalil 2002, Thm 4.1]. ∎

Here $-\ln O(y) = -\ln(1 - q_{\text{est}}^2) - (5/2)q_{\text{est}}^2$ is not convex (its second q_{est}-derivative at 0 is $2 - 5 < 0$), and the superlevel sets $\{O \geq c\}$ for $1 < c < (2/5)e^{3/2}$ are not convex ($O(0) = 1$, while O is largest, $(2/5)e^{3/2}$, at $q_{\text{est}}^2 = 3/5$). Condition 3 of Thm 5.1 constrains only the zone $\{O \geq 0\}$.

En llano: ejemplo comprobable a mano: con el error cuadrático de GTPA y una barrera logarítmica sobre una zona segura convexa hay dos mínimos igual de buenos, así que no hay un único punto de llegada.

CT-04 — The three conditions of Thm 5.1 are not necessary [PROVED_HERE] (EG-03.1, Thm 5.1)

(a) Condition 1 (G_{model} strictly positive definite) is not necessary. $d = 2$, $y = (q, p) \coloneqq (q_{\text{est}}, p_{\text{est}}) \in R^2 \times R^2$, $G_{\text{model}} = \text{diag}(1, 0)$ (singular), $M_{\text{cog}} = I$, $y^* = 0$, $\mu = 1$, $O(y) \coloneqq 1 - |q|^2$ (concave; $U = \{|q| < 1\} \times R^2$). Then

$$V(y) = \tfrac{1}{2}\, q_1^2 - \ln(1 - |q|^2) + \tfrac{1}{2}\, |p|^2$$

is strictly convex on U with compact sublevel sets, so by CT-02 (a)–(c) it has exactly one critical point, $y = 0$, which attracts every solution of both flows.

Proof. The q-Hessian of $-\ln(1 - |q|^2)$ is $[2(1 - |q|^2)I + 4qq^{\mathsf{T}}]/(1 - |q|^2)^2 \succ 0$ for $|q| < 1$; $q_1^2/2$ is convex; the p-part has Hessian I. $V \geq \tfrac{1}{2}|p|^2$ (both other terms are ≥ 0) and $-\ln(1 - |q|^2) \to \infty$ as $|q| \to 1$, so sublevel sets are bounded and closed. $\nabla_q V = (q_1 + 2q_1/(1 - |q|^2), 2q_2/(1 - |q|^2))$ and $\nabla_p V = p$ vanish only at $y = 0$. ∎

(b) Condition 3 (convex safe zone) is not necessary. $k = 2$ (a model state $y \in R^2$; a p_{est}-block with its own quadratic term would add a separate factor with a unique minimizer), $K = I$, $y^* = 0$, $\mu = 1$, and $O \coloneqq \text{dist}_C \coloneqq \text{dist}(\cdot, C)$, the distance to the closed quadrant $C \coloneqq [1, \infty)^2$ — v1.2's "distance to a cliff". The safe zone $U \coloneqq R^2 \setminus C$ is not convex ($(2, 0)$ and $(0, 2)$ lie in U, their midpoint $(1, 1)$ in C). Then $V(y) \coloneqq \tfrac{1}{2}|y|^2 - \ln \text{dist}_C(y)$ has exactly one critical point on U, (s^*, s^*) with $s^* \coloneqq (1 - \sqrt{3})/2 \approx -0.366$; it is the unique minimizer, and every solution of $\dot{y} = -\nabla V(y)$ started in U converges to it.

Proof. On U, $\mathrm{dist}_C(y) = |y-(1,1)|$ if $y_1 < 1$ and $y_2 < 1$; $\mathrm{dist}_C(y) = 1 - y_2$ if $y_1 \geq 1$ (then $y_2 < 1$); $\mathrm{dist}_C(y) = 1 - y_1$ if $y_2 \geq 1$ (then $y_1 < 1$). The three formulas and their first derivatives agree on the two interfaces (on $\{y_1 = 1, y_2 < 1\}$ both give the value $1 - y_2$ and the gradient $(0, -1)$; symmetrically on the other), so $\mathrm{dist}_C \in C^1(U)$ with a locally Lipschitz gradient, and so is V. $\nabla V = y - \nabla\,\mathrm{dist}_C / \mathrm{dist}_C$. In the second region $\partial V/\partial y_1 = y_1 \geq 1$, in the third $\partial V/\partial y_2 = y_2 \geq 1$: no critical points there. In the first, with $d_1 := |y-(1,1)|$, $\nabla V = y - (y-(1,1))/d_1^2$ vanishes iff $y_i(d_1^2 - 1) = -1$ for $i = 1, 2$; so $d_1^2 \neq 1$ and $y_1 = y_2 =: s = -1/(d_1^2 - 1)$. If $d_1^2 < 1$ then $s = 1/(1 - d_1^2) > 1$, impossible in this region; so $d_1^2 > 1$ and $s < 0$, and $d_1^2 = 2(1-s)^2$ turns $s(d_1^2-1) = -1$ into $2s(1-s)^2 - s + 1 = (s-1)(2s^2 - 2s - 1) = 0$, whose roots are 1 and $(1 \pm \sqrt{3})/2$. The only negative one is $s^* = (1-\sqrt{3})/2$, and there $d_1^2 = 2(1-s^*)^2 > 2 > 1$ as required. Since $\mathrm{dist}_C(y) \leq |y-(1,1)| \leq |y| + \sqrt{2}$, $V(y) \geq \tfrac{1}{2}|y|^2 - \ln(|y| + \sqrt{2}) \to \infty$ as $|y| \to \infty$, and $V \to \infty$ at ∂U ($\mathrm{dist}_C \to 0$); so the sublevel sets of V are compact subsets of U, V attains its minimum at a critical point, hence at (s^*, s^*). The flow: the proof of CT-02 (b) uses only that ∇V is locally Lipschitz, that the sublevel sets are compact and that (s^*, s^*) is the only critical point. ∎

(c) Condition 2 (M_{cog} positive definite) holds automatically for the object of Def 1.1 (B-07 (b)); it cannot separate landscapes with one attractor from landscapes with several.

En llano: las tres condiciones de GTPA no son necesarias: puede haber un único punto de llegada con una métrica que no es estrictamente positiva, o con una zona segura que no es convexa (por ejemplo, con la distancia a un "acantilado" en forma de esquina).

CT-05 — The quadratic control problem of Def 7.1–7.2 [PROVED_HERE] (EG-03.8, Thm 7.1)

Integers $k, m \geq 1$ (k controls; m components of v1.2's F_{req}); $B \in R^{m\times k}$; $G \in R^{m\times m}$ symmetric (v1.2's G_{reg}); $F \in R^m$ (F_{req}); $\lambda \in R$ ($\lambda_{\mathrm{effort}}$). Let

$$A(u) := \tfrac{1}{2}\langle F - Bu, G(F - Bu)\rangle + \tfrac{1}{2}\lambda\,|u|^2 \qquad (\mathrm{Action}_{\mathrm{Local}},\ p.43)$$
$$Z \;:= B^{\mathsf{T}}GB + \lambda I \qquad (\mathrm{Def}\ 7.2)$$
$$b \;:= B^{\mathsf{T}}GF.$$

(a) $\nabla A(u) = Zu - b$ and $D^2A = Z$.

(b) Z is singular iff $-\lambda$ is an eigenvalue of $B^{\mathsf{T}}GB$. If $G \succeq 0$ (so $B^{\mathsf{T}}GB \succeq 0$): for $\lambda > 0$, $Z \succeq \lambda I$ is invertible whatever the rank of B; for $\lambda = 0$ and $G \succ 0$, Z is singular iff $\ker B \neq \{0\}$; for $\lambda < 0$, iff $-\lambda \in \mathrm{spec}(B^{\mathsf{T}}GB)$. If G is not positive semidefinite, Z can be singular with $\lambda > 0$ ($k = m = 1$, $B = 1$, $G = -\lambda$).

(c) If $G \succeq 0$ and $\lambda > 0$: A is strictly convex, its unique minimizer is $u^* = Z^{-1}b$, $|u^*| \leq |b|/\lambda$, and $\langle v, u^*\rangle = 0$ for every $v \in \ker B$.

(d) If $G \succeq 0$ and $\lambda = 0$: the minimizers of A are the solutions of $Zu = b$, a nonempty affine set, and the one of least norm is Z^+b.

(e) The printed algorithm with threshold $\tau > 0$ (v1.2's "epsilon"): with the decomposition $Z = W\,S\,W^{\mathsf{T}}$ of §1, step 3 returns

$$u_\tau := \sum_{\sigma_i > \tau} \sigma_i^{-1}\,\langle w_i, b\rangle\, w_i$$

(the same vector for every orthonormal eigenbasis: it is $\varphi(Z)b$ with $\varphi(\sigma) = \sigma^{-1}$ for $\sigma > \tau$ and 0 otherwise). If $G \succeq 0$ and $\lambda > 0$: $u_\tau = u^*$ iff $\langle w_i, b\rangle = 0$ for every i with $\sigma_i \leq \tau$ — in particular whenever $\tau < \lambda$, since every $\sigma_i \geq \lambda$; otherwise $A(u_\tau) - A(u^*) = \tfrac{1}{2}\langle u_\tau - u^*, Z(u_\tau - u^*)\rangle > 0$. Example: $k = m = 1$, $B = \delta \in (0, 1)$, $G = 1$, $\lambda = 1$, $F = 1$, $\tau = 2$: $Z = 1 + \delta^2 < \tau$, so $u_\tau = 0$, while $u^* = \delta/(1+\delta^2)$ and $A(u^*) = 1/(2(1+\delta^2)) < \tfrac{1}{2} = A(0)$.

(f) The printed formula `u_opt = Z^(-1) * ( Gradient_u( Action_Local ) )` depends on the point where the gradient is taken: at $u = 0$ it gives $-Z^{-1}b = -u^*$. From any u_0, the step $u_0 - Z^{-1}\nabla A(u_0)$ equals $Z^{-1}b$ (A is quadratic).

Proofs. (a) Expand A. (b) $Zv = 0$ iff $B^\mathsf{T}GBv = -\lambda v$. If $G \succeq 0$, $\langle v, B^\mathsf{T}GBv\rangle = |G^{1/2}Bv|^2 \geq 0$, so $Z \succeq \lambda I$; for $\lambda = 0$ and $G \succ 0$, $B^\mathsf{T}GBv = 0$ implies $|G^{1/2}Bv|^2 = 0$, i.e. $Bv = 0$. (c) $D^2A = Z \succeq \lambda I \succ 0$; the critical point solves $Zu = b$; $|u^*| \leq \|Z^{-1}\|\,|b| \leq |b|/\lambda$. For $v \in \ker B$, $Zv = \lambda v$, so $Z^{-1}v = v/\lambda$ and $\langle v, u^*\rangle = \langle Z^{-1}v, b\rangle = \lambda^{-1}\langle Bv, GF\rangle = 0$. (d) With $N \coloneqq G^{1/2}B$, $A(u) = \tfrac{1}{2}|G^{1/2}F - Nu|^2$ is convex, so its minimizers are its critical points, the solutions of $N^\mathsf{T}Nu = N^\mathsf{T}G^{1/2}F$, i.e. $Zu = b$. Since $\ker(N^\mathsf{T}N) = \ker N$, $\operatorname{range} Z = (\ker Z)^\perp = (\ker N)^\perp = \operatorname{range} N^\mathsf{T} \ni b$: solutions exist, and they are $Z^+b + \ker Z$ (ZZ^+ is the orthogonal projection onto range Z, and $Z^+b \in \operatorname{range} Z$). For $n \in \ker Z$, $|Z^+b + n|^2 = |Z^+b|^2 + |n|^2$. (e) $u^* = \sum_i \sigma_i^{-1}\langle w_i, b\rangle w_i$, so $u_\tau - u^* = -\sum_{\sigma_i \leq \tau} \sigma_i^{-1}\langle w_i, b\rangle w_i$; A is quadratic with $\nabla A(u^*) = 0$, so $A(u) - A(u^*) = \tfrac{1}{2}\langle u - u^*, Z(u - u^*)\rangle \geq \tfrac{1}{2}\lambda|u - u^*|^2$. The example: $A(u) = \tfrac{1}{2}(1 - \delta u)^2 + \tfrac{1}{2}u^2$. (f) Direct. ∎

En llano: con un coste de esfuerzo positivo, el problema de control de GTPA siempre tiene una única mejor acción, finita, aunque haya direcciones en las que no se puede actuar (en ellas no empuja); el filtro del algoritmo solo da esa mejor acción si su umbral es pequeño; si no, da otra peor.

CT-06 — Quadratic cost: the control is zero at most at one instant `[DECLARED model + PROVED_HERE + STANDARD_FACT]` (EG-03.9, Thm 7.2)

Declared model: the simplest (q,p) setting of v1.2's Ch. 7 (a separate object from the base motion, U-11): $\ddot{q} = u$ on $[0, T]$, $q \in R$, from rest to rest over a distance $D > 0$: $q(0) = \dot{q}(0) = 0$, $q(T) = D$, $\dot{q}(T) = 0$; controls $u \in L^2(0, T)$. Cost

$$\mathrm{Cost}_c(u) \coloneqq \int_0^T (\tfrac{1}{2}u^2 + \tfrac{1}{2}c\,\dot{q}^2)\,dt, \qquad c \geq 0 :$$

a quadratic effort term — v1.2's "|| B(q) * u ||^2" (p.30) with $B = 1$ — plus an optional quadratic term in $\dot{q}$; the landscape is flat, so a term for "work done against J" is 0.

Then Cost_c has exactly one minimizer u^* among admissible controls, and u^* vanishes at most at one instant: for $c = 0$, $u^*(t) = (6D/T^2)(1 - 2t/T)$; for $c > 0$, $u^*(t) = a_1\cosh(\sqrt{c}\,t) + a_2\sinh(\sqrt{c}\,t)$ with $(a_1, a_2) \neq (0, 0)$. In particular u^* vanishes on no interval: the minimizer has no phase with $u = 0$.

Proof. Write $\langle\cdot,\cdot\rangle$ for the $L^2(0, T)$ product, $v \coloneqq \dot{q} = Vu$ with $(Vu)(t) \coloneqq \int_0^t u$, and $(V^*f)(t) = \int_t^T f$ for its adjoint. The end conditions are $\langle u, 1\rangle = 0$ ($v(T) = 0$) and $\langle u, T - t\rangle = D$ ($q(T) = \int_0^T (T - t)u(t)\,dt$); they define a closed affine subspace $\mathcal{A} = u_1 + \mathcal{A}_0$ of L^2, nonempty (it contains the $c = 0$ formula), with $\mathcal{A}_0 = \{1, T - t\}^\perp$. $\mathrm{Cost}_c(u) = \tfrac{1}{2}\langle u, (I + cV^*V)u\rangle$, and $\langle u, w\rangle_c \coloneqq \langle u, (I + cV^*V)w\rangle$ is an inner product equivalent to $\langle\cdot,\cdot\rangle$ ($\langle u, u\rangle \leq \langle u, u\rangle_c \leq (1 + c\|V\|^2)\langle u, u\rangle$). So minimizing Cost_c over $\mathcal{A}$ is finding the point of $\mathcal{A}$ nearest to 0 for $\langle\cdot,\cdot\rangle_c$: it exists and is unique, and it is characterized by $\langle u^*, w\rangle_c = 0$ for all $w \in \mathcal{A}_0$ [`STANDARD_FACT`: projection onto a nonempty closed convex set of a Hilbert space, Brezis 2011, Thm 5.2]. That is, $(I + cV^*V)u^* \in \mathcal{A}_0^\perp = \operatorname{span}\{1, T - t\}$:

$$u^*(t) = \alpha + \beta\,(T - t) - c\int_t^T v^*(s)\,ds, \qquad v^* \coloneqq Vu^*.$$

The right side is continuous, so u^* is continuous, $v^* \in C^1$, the right side is C^1, and $u^{*\prime} = -\beta + c\,v^*$, $u^{*\prime\prime} = c\,u^*$. For $c = 0$, $u^* = \alpha + \beta(T - t)$, and the end conditions $\alpha T + \beta T^2/2 = 0$, $\alpha T^2/2 + \beta T^3/3 = D$ give $\beta = 12D/T^3$, $\alpha = -6D/T^2$, i.e. $u^*(t) = (6D/T^2)(1 - 2t/T)$, zero only at $t = T/2$. For $c > 0$, $u^* = a_1\cosh(\sqrt{c}\,t) + a_2\sinh(\sqrt{c}\,t)$; $u^* \not\equiv 0$ because $D > 0$; if $a_2 = 0$ it has no zero, and if $a_2 \neq 0$ it vanishes iff $\tanh(\sqrt{c}\,t) = -a_1/a_2$, which has at most one solution. ∎

En llano: si el esfuerzo se paga al cuadrado, como en la lista de costes de GTPA, la mejor forma de llevar algo de un reposo a otro empuja y luego empuja en sentido contrario de forma continua, sin

ningún tramo en que no se haga nada.

CT-07 — L^1 cost with a bound: push, zero, reverse `[DECLARED model + PROVED_HERE]` (EG-03.9, Thm 7.2)

Declared model of CT-06, with measurable controls $|u(t)| \le u_{\max}$ a.e. ($u_{\max} > 0$) and cost $\mathrm{Cost}_1(u) \coloneqq \int_0^T |u|\,dt$.

(a) An admissible control exists iff $T \ge T_{\min} \coloneqq 2(D/u_{\max})^{1/2}$.

(b) For $T \ge T_{\min}$, the minimum of Cost_1 is $2V^*$, with $V^* \coloneqq (u_{\max}\,T - (u_{\max}^2\,T^2 - 4\,u_{\max}\,D)^{1/2})/2$, and it is attained exactly (up to null sets) by

$$u = +u_{\max} \text{ on } (0, t_1), \qquad u = 0 \text{ on } (t_1, T - t_1), \qquad u = -u_{\max} \text{ on } (T - t_1, T),$$
$$t_1 \coloneqq V^*/u_{\max}.$$

The middle interval has length $(T^2 - 4D/u_{\max})^{1/2}$, positive iff $T > T_{\min}$. So the three phases of Thm 7.2 (v1.2's "Pulse", "Coast", "Brake") are exactly the unique minimizer here, for every $T > T_{\min}$; at $T = T_{\min}$ the middle phase disappears.

Proof. Let $v \coloneqq \dot q$: absolutely continuous, $\dot v = u$ a.e., $v(0) = v(T) = 0$, $\int_0^T v = D$. Since $|\dot v| \le u_{\max}$, $v(t) \le \min(u_{\max}\,t, u_{\max}(T - t))$; with $m \coloneqq \max v$ (positive, as $\int v = D > 0$), $v \le \varphi_m \coloneqq \min(m, u_{\max}\,t, u_{\max}(T - t))$ and $D \le \Phi(m) \coloneqq \int_0^T \varphi_m$. Φ is non-decreasing, with $\Phi(m) = mT - m^2/u_{\max}$ for $0 \le m \le u_{\max}\,T/2$ (the area of a trapezoid) and $\Phi(m) = u_{\max}\,T^2/4$ for $m \ge u_{\max}\,T/2$. (a) So $D \le u_{\max}\,T^2/4$, i.e. $T \ge T_{\min}$; conversely, for $T \ge T_{\min}$ the control of (b) is admissible. (b) V^* is the smaller root of $\Phi(m) = D$; Φ is strictly increasing on $[0, u_{\max}\,T/2]$ and $V^* \le u_{\max}\,T/2$, so $\Phi(m) \ge D$ forces $m \ge V^*$. For a maximizer t_m of v, $\mathrm{Cost}_1(u) = \int |\dot v| \ge |v(t_m) - v(0)| + |v(T) - v(t_m)| = 2m \ge 2V^*$. The control of (b) gives $v = \varphi_{V^*}$, $\int v = \Phi(V^*) = D$ and $\mathrm{Cost}_1 = 2u_{\max}\,t_1 = 2V^*$. If $\mathrm{Cost}_1(u) = 2V^*$, then $m = V^*$ and $\int v = D = \int \varphi_{V^*}$ with $v \le \varphi_{V^*}$, both continuous, so $v = \varphi_{V^*}$ and $u = \dot v$ is the control of (b) a.e. The middle length is $T - 2V^*/u_{\max} = (T^2 - 4D/u_{\max})^{1/2}$. ∎

En llano: si el esfuerzo se paga por su tamaño (no al cuadrado) y tiene un tope, la mejor forma es empujar al máximo, dejar de empujar y luego empujar al máximo en sentido contrario, y es la única mejor forma cuando hay tiempo de sobra.

Remark (sufficient, not necessary). The cost $\int |u|$ with a cap is one condition that gives the three phases; it is not needed. Same model with no cap, $T = 4$, $D = 1$ and cost $\int (|u| + 0.05u^2)$: the problem is convex with linear dynamics, so the extremal of the minimum principle is the minimizer, unique because $0.05u^2$ is strictly convex. It is the soft threshold of an affine function of t (minus the costate of $\dot q$), by symmetry $p(t) = k(2 - t)$: $u = \mathrm{sign}(p) \cdot \max(|p| - 1, 0)/0.1$. So u falls linearly from its peak to 0 on $(0, t_1)$, $u = 0$ on $(t_1, 4 - t_1)$ and the mirror image brakes on $(4 - t_1, 4)$, with $t_1 = 2 - 1/k$ the root in $(0, 2)$ of $10t_1^2(6 - t_1) = 3(2 - t_1)$. Numbers: $t_1 \approx 0.299$, a zero phase of length ≈ 3.40 (85% of the horizon), peak $|u| \approx 1.76$, cost ≈ 0.557. Recomputed by machine on 2026-10-05 (run once, nothing added to the repository): a direct discretization (800 steps, SciPy SLSQP) pushes on $(0, 0.300)$, holds $|u| < 10^{-4}$ for 3.400, brakes on $(3.700, 4)$, cost 0.5571, as the closed form (0.5571). Level 2. With the quadratic part alone there is no zero phase (CT-06).

CT-08 — Gradient fields have no curl; the decrease of Ψ needs none `[PROVED_HERE + STANDARD_FACT]` (EG-03.10, Thm 8.1)

(a) If $V \in C^2(\Omega)$ on an open $\Omega \subset R^n$, then $\mathrm{curl}(-\nabla V) \equiv 0$ ($\partial_i \partial_j V = \partial_j \partial_i V$). `STANDARD_FACT` (converse): on an open set that is star-shaped with respect to a point, a C^1 field with zero curl is a gradient [Poincaré lemma: Spivak 1965, Thm 4-11, for the closed 1-form $\sum_i F_i\,dx_i$].

(b) Let $V \in C^2(R^n)$ and $\zeta > 0$. Along every solution of $\ddot{x} = -\nabla V(x) - \zeta\dot{x}$, the function $\Psi(x,\dot{x}) \coloneqq \tfrac{1}{2}|\dot{x}|^2 + V(x)$ satisfies $d\Psi/dt = -\zeta|\dot{x}|^2 \le 0$. So a solution that starts in $\{\Psi < h\}$ stays in it for all later times: leaving a sublevel set of Ψ (for instance the region around a local minimizer of V cut out by a level of Ψ) needs an added control. The part of the right side that depends on x only, $-\nabla V$, has zero curl by (a); the term $-\zeta\dot{x}$ depends on $\dot{x}$, so it is not a vector field on the state space, and "its curl on Q" is not defined.

(c) Curl of the base drift (U-01, noiseless). For constant A, G and $J \in C^2$ with $H \coloneqq D^2J$,

$$\operatorname{curl}[(A-G)\nabla J] = (GH - HG) - (AH + HA).$$

Examples, $n = 2$, with the rotation matrix $S = [[0,1],[-1,0]]$ of U-03 (b): $A = S$, $G = 0$, $J = |q|^2/2$ gives $(\operatorname{curl})_{12} = -2$; $A = 0$, $G = \operatorname{diag}(1,2)$, $J = q_1q_2$ gives $(\operatorname{curl})_{12} = -1$; $A = 0$, $G = gI$ gives 0 for every J. For constant $G \succ 0$, $G^{-1}(-G\nabla J) = -\nabla J$: the sliding part is a gradient for the inner product $\langle\cdot,\cdot\rangle_{G^{-1}}$, so whether a field "has curl" also depends on the inner product used to turn vectors into covectors.

Proof. (b) $d\Psi/dt = \langle\dot{x}, -\nabla V - \zeta\dot{x}\rangle + \langle\nabla V, \dot{x}\rangle = -\zeta|\dot{x}|^2$. (c) $F \coloneqq (A-G)\nabla J$ has $\partial_j F_i = ((A-G)H)_{ij}$, so $(\operatorname{curl} F)_{ij} = ((A-G)H)_{ji} - ((A-G)H)_{ij}$, i.e. $\operatorname{curl} F = H(A-G)^{\mathsf{T}} - (A-G)H = -HA - HG - AH + GH$ ($A^{\mathsf{T}} = -A$, $G^{\mathsf{T}} = G$, $H^{\mathsf{T}} = H$). Examples: $-2S$ has $(1,2)$ entry -2; $GH - HG = [[0,-1],[1,0]]$ for the second; $gH - Hg = 0$. ∎

En llano: un campo que es la pendiente de un paisaje no tiene rotacional, se pierda o no "altura" por el camino; perder altura por un término que depende de la rapidez del estado hace que cueste salir de un pozo, pero no crea rotacional; el rotacional de la deriva de UFD sale de la parte que gira o de una movilidad distinta según la dirección.

CT-09 — Individual effort at a fixed total `[PROVED_HERE]` (EG-03.0, Thm 9.1)

Let $\langle\cdot,\cdot\rangle$ be an inner product on R^d (for instance $\langle\cdot,\cdot\rangle_{G_{\text{reg}}}$ with $G_{\text{reg}} \succ 0$), $|\cdot|$ its norm, $N \ge 2$, $v_1, \ldots, v_N \in R^d$, $F \coloneqq \sum_i v_i$, $S_\times \coloneqq \sum_{i<j}\langle v_i, v_j\rangle$, and, for nonzero v_i, v_j, $C_{ij} \coloneqq \langle v_i, v_j\rangle/(|v_i||v_j|)$ (Def 9.1).

(a) $\sum_i |v_i|^2 = |F|^2 - 2S_\times$, and $S_\times = \sum_{i<j} C_{ij}\,|v_i||v_j|$ when every $v_i \neq 0$.

(b) For a fixed total F, $\sum_i |v_i|^2 \ge |F|^2/N$, with equality iff $v_i = F/N$ for every i (then, if $F \neq 0$, $C_{ij} = 1$ for all pairs).

(c) For fixed norms $|v_i| = r_i > 0$, $|F|^2 = \sum_i r_i^2 + 2\sum_{i<j} C_{ij}\,r_i\,r_j$. For $N = 2$ and $r_1 = r_2 = 1$, $|F|^2 = 2 + 2C_{12}$: at fixed individual norms, the squared total grows with coherence.

Proof. (a) Expand $|\sum_i v_i|^2$. (b) $\sum_i |v_i - F/N|^2 = \sum_i |v_i|^2 - |F|^2/N \ge 0$. (c) is (a). ∎

En llano: con el total fijo, cuanto más alineados están los vectores menos esfuerzo individual hace falta (la suma de los cuadrados), y el mínimo es con todos iguales; con los esfuerzos individuales fijos, en cambio, el cuadrado del total crece con la alineación.

CT-10 — Full coherence: one ray and one blind spot `[DECLARED model + PROVED_HERE]` (EG-03.11, EG-03.2, Thm 9.2)

Setting of CT-09, every $v_i \neq 0$. Declared response model: the collective can produce every vector of $\text{Reach} \coloneqq \{\sum_i a_i\,v_i \,:\, a_i \ge 0\}$ (each agent scales its own intent and keeps its direction); the residual of a perturbation $w \in R^d$ is $r(w) \coloneqq \inf_{f\in\text{Reach}} |w + f|$.

(a) $C_{ij} = 1$ for all pairs iff $v_i = |v_i|\,e$ for every i, for one unit vector e.

(b) Then Reach $= \{te : t \geq 0\}$ and $r(w) \geq |w_\perp|$, with $w_\perp := w - \langle w, e\rangle e$, and equality iff $\langle w, e\rangle \leq 0$. Every perturbation with $w_\perp \neq 0$ leaves a residual of at least $|w_\perp| > 0$, whatever the collective does: $e^\perp$ (of dimension $d-1$) is the shared blind spot.

(c) If Reach $= R^d$ (the intents positively span R^d), then $r \equiv 0$, and some pair has $C_{ij} < 0$. Example ($d = 2$, Euclidean): $v_1 = e_1$, $v_2 = e_2$, $v_3 = -(e_1 + e_2)/\sqrt{2}$ give Reach $= R^2$, with $C_{13} = C_{23} = -1/\sqrt{2}$; the orthogonal pair e_1, e_2 alone gives the closed quadrant, and $r((1,1)) = \sqrt{2}$.

(d) Declared motion $\dot{x} = f(t) + w$, with $f(t) \in$ Reach locally integrable and w constant. In case (b), $x_\perp(t) = x_\perp(0) + t\, w_\perp$ for every $t \geq 0$: the component orthogonal to e moves at the constant rate $|w_\perp|$ whatever the collective does, and nothing in the model changes the v_i or the C_{ij}.

(e) If w is random with a density on R^d, $d \geq 2$, the probability of full correction Prob$(r(w) = 0)$ is 0 in case (b) and 1 in case (c). (If the scalar coherence of a collective is declared as the mean of its C_{ij} — the summary v1.2 uses in §9.4, pp.53–54 — it equals 1 iff every $C_{ij} = 1$, since each $C_{ij} \leq 1$.)

Proofs. (a) Cauchy–Schwarz: $\langle v_i, v_j\rangle = |v_i||v_j|$ iff v_j is a positive multiple of v_i. (b) $|w + te|^2 = |w_\perp|^2 + (\langle w, e\rangle + t)^2$, minimized over $t \geq 0$ at $t = \max(0, -\langle w, e\rangle)$. (c) With $s := \sum_i v_i$, $-s \in$ Reach gives $\sum_i c_i\, v_i = 0$ with $c_i = a_i + 1 > 0$; then $0 = |\sum_i c_i\, v_i|^2 = \sum_i c_i^2\, |v_i|^2 + 2\, \sum_{i<j} c_i\, c_j\, \langle v_i, v_j\rangle$, impossible if every $\langle v_i, v_j\rangle \geq 0$. Example: for $(\alpha_1, \alpha_2) \in R^2$ take $a_3 := \sqrt{2}\, \max(0, -\alpha_1, -\alpha_2)$, $a_1 := \alpha_1 + a_3/\sqrt{2} \geq 0$, $a_2 := \alpha_2 + a_3/\sqrt{2} \geq 0$; and $\inf_{a_1, a_2 \geq 0} |(1 + a_1, 1 + a_2)| = \sqrt{2}$. (d) Project the equation on $e^\perp$: $f(t) \in$ Re. (e) By (b), $r(w) = 0$ iff $w \in \{-te : t \geq 0\}$, a half-line, of measure 0 for $d \geq 2$. ∎

En llano: si todos los vectores apuntan igual, el colectivo solo puede empujar en una dirección y no puede corregir nada que venga de lado (ese es su punto ciego); para poder corregirlo todo hacen falta algunos que apunten en sentidos opuestos; y un empujón lateral no "rompe" nada al instante: deja un desvío que crece a ritmo constante.

CT-11 — Where the maximizers of Efficiency × Resilience lie `[PROVED_HERE]` (EG-03.2, Thm 9.3)

Let $c_0 < 1$, Eff, Res $: [c_0, 1] \to [0, \infty)$ and $P := \text{Eff} \cdot \text{Res}$.

(a) If P is upper semicontinuous, argmax P is nonempty.

(b) If moreover Res$(1) = 0$ and $P(c) > 0$ for some c, every maximizer is < 1. No monotonicity of `Eff` is used.

(c) Res$(1) = 0$ cannot be replaced by "`Res` decreasing": on $[0, 1]$, Eff$(c) = e^{2c}$ and Res$(c) = 2 - c$ give $P'(c) = (3 - 2c)e^{2c} > 0$, so argmax $P = \{1\}$.

(d) Without upper semicontinuity argmax P can be empty: on $[0, 1]$, Eff $\equiv 1$, Res$(c) = c$ for $c < 1$ and Res$(1) = 0$; $\sup P = 1$ is not attained.

(e) The maximizer need not be unique: on $[0, 1]$, Eff $\equiv 1$, Res$(c) = \min(1, 2 - 2c)$ give argmax $P = [0,$ ½$]$.

Proofs. (a) P is bounded above: otherwise $P(c_j) \to \infty$ along a sequence with a convergent subsequence $c_j \to \bar{c}$, and upper semicontinuity gives $P(\bar{c}) \geq \limsup P(c_j) = \infty$. Take c_j with $P(c_j) \to \sup P$ and a convergent subsequence $c_j \to \bar{c}$: $P(\bar{c}) \geq \limsup P(c_j) = \sup P$. (b) $P(1) = \text{Eff}(1) \cdot 0 = 0 < \max P$. (c)–(e) Direct. ∎

En llano: si la resiliencia vale cero con coherencia total y el producto no es cero en todas partes, el óptimo (si existe) está por debajo de la coherencia total; sin ese cero puede estar justo en la coherencia total, puede no existir si la función salta y puede no ser único.

CT-12 — A Class A model: collapse from one open set, recovery from another `[DECLARED model + PROVED_HERE]` (EG-03.3, Thm 10.1)

Declared model $\mathfrak{A}$ (all scalars; $a \in R$, $\lambda, \beta, \gamma > 0$, $k \coloneqq \beta/\lambda$):

- territory: $\dot{x} = ax + u$, $x \in R$, goal $x = 0$ (for $a > 0$, without action the territory runs away from its goal);
- perception $\Pi_s(x) \coloneqq sx$ with $s > 0$ (a linear diffeomorphism of R, so Ax 2.2 holds); constitution: a weight $\theta > 0$ on the perceived error;
- $E_{\text{model}} \coloneqq \tfrac{1}{2}\theta^2 s^2 x^2$ (the perceived, weighted error) and $E_{\text{base}} \coloneqq \tfrac{1}{2}x^2$ (the territory's error);
- cost rate (the integrand of S_{total}): $\ell \coloneqq \tfrac{1}{2}\lambda u^2 + \beta E_{\text{model}}$;
- update law, a myopic descent of S_{total}: $u \coloneqq \operatorname{argmin}_u[\tfrac{1}{2}\lambda u^2 + \beta\, \partial_x E_{\text{model}} \cdot (ax + u)] = -k\theta^2 s^2 x$, $\dot{s} \coloneqq -\gamma\, \partial_s E_{\text{model}} = -\gamma\theta^2 s x^2$, $\dot{\theta} \coloneqq -\gamma\, \partial_\theta E_{\text{model}} = -\gamma\theta s^2 x^2$.

Put $r \coloneqq \theta s$. Along solutions $s^2 - \theta^2$ is constant, $\dot{r} = -\gamma r(s^2 + \theta^2)x^2 \le -2\gamma r^2 x^2$, $\dot{x} = (a - kr^2)x$ and $E_{\text{model}} = \tfrac{1}{2}r^2 x^2$.

(a) Well posed, and Class A holds literally: for $x_0 \neq 0$ and $s_0, \theta_0 > 0$ the solution exists for all $t \ge 0$, $s(t), \theta(t) > 0$, $x(t) \neq 0$, and $\dot{s}(t) < 0$, $\dot{\theta}(t) < 0$ for every t (Def 10.1: $||d\Theta/dt|| > 0$, and the perception is moved freely).

(b) Collapse. If $a > 0$, $x_0 \neq 0$ and $k(s_0\theta_0)^2 < a$, then $E_{\text{base}}(t) \to \infty$ and $E_{\text{model}}(t) \to 0$.

(c) Recovery. Let $a > 0$ and $r_1 \coloneqq (2a/k)^{1/2}$. If $s_0\theta_0 > r_1$ and $\gamma(|s_0^2 - \theta_0^2| + 2s_0\theta_0)x_0^2 < 2a\,\ln(s_0\theta_0/r_1)$, then $r(t) > r_1$ for all t, $E_{\text{base}}(t) \le E_{\text{base}}(0)e^{-2at} \to 0$ and $E_{\text{model}}(t) \to 0$.

Both sets of initial conditions are open and nonempty in $\{x_0 \neq 0, s_0 > 0, \theta_0 > 0\}$.

(d) If $a \le 0$, E_{base} is non-increasing, from every start.

(e) Maximizing instead: $\sup_u \ell = +\infty$ at every state ($\lambda > 0$). More generally, let a cost rate at a fixed state be $|Bu|^2 + m(u)$ with $B \neq 0$ (as v1.2's effort term $||B(q)u||^2$, p.30) and $m(u) \ge -C(1 + |u|)$ for some constant C (the other terms bounded below, or at most linear in u); then its supremum over u is $+\infty$, and an agent "maximizing" it has no action to choose. v1.2 gives no formulas for its other terms (p.30: kinetic cost, work against J, dissonance, complexity, erasure cost); read as nonnegative, except the work term read as linear in u, they satisfy this condition.

Proofs. The right side of the equations is polynomial. $s(t) = s_0 \exp(-\gamma \int_0^t \theta^2 x^2) > 0$ and $\theta(t) = \theta_0 \exp(-\gamma \int_0^t s^2 x^2) > 0$, both non-increasing, so $r \le r_0 \coloneqq s_0\theta_0$ and $|\dot{x}| \le (|a| + kr_0^2)|x|$: no blow-up, the solution is global, and $x(t) = x_0 \exp(\int_0^t (a - kr^2)) \neq 0$. $d/dt(s^2 - \theta^2) = -2\gamma\theta^2 s^2 x^2 + 2\gamma\theta^2 s^2 x^2 = 0$; $\dot{r} = \dot{\theta}s + \theta\dot{s} = -\gamma r(s^2 + \theta^2)x^2$, and $s^2 + \theta^2 \ge 2s\theta$. This gives (a). (b) $kr^2 \le kr_0^2 < a$ gives $\dot{x}/x \ge \delta \coloneqq a - kr_0^2 > 0$, so $|x(t)| \ge |x_0|e^{\delta t}$ and $E_{\text{base}} \to \infty$. From $\dot{r} \le -2\gamma r^2 x^2$, $1/r(t) \ge 2\gamma \int_0^t x^2$; from $\dot{x}/x \le a$, $|x(\tau)| \ge |x(t)|e^{-a(t-\tau)}$ for $\tau \le t$, so $\int_0^t x^2 \ge x(t)^2(1 - e^{-2at})/(2a)$. Hence $E_{\text{model}}(t) = \tfrac{1}{2}r^2x^2 \le a^2/(2\gamma^2 x(t)^2(1 - e^{-2at})^2) \to 0$. (c) $(s^2 + \theta^2)^2 = (s^2 - \theta^2)^2 + 4r^2$, so $s^2 + \theta^2 \le |s_0^2 - \theta_0^2| + 2r_0 =: \kappa$ and $d(\ln r)/dt \ge -\gamma\kappa x^2$. While $r \ge r_1$ on $[0, t]$, $\dot{x}/x \le a - 2a = -a$, so $|x(\tau)| \le |x_0|e^{-a\tau}$ and $\ln r(\tau) \ge \ln r_0 - \gamma\kappa x_0^2/(2a) > \ln r_1$ by hypothesis. Let $t^* \coloneqq \sup\{t : r \ge r_1 \text{ on } [0, t]\} > 0$; if $t^* < \infty$, then $r(t^*) > r_1$ by the bound and $r > r_1$ a little beyond t^* by continuity, a contradiction. So $r > r_1$ for all t, $|x(t)| \le |x_0|e^{-at}$, and $E_{\text{model}} = \tfrac{1}{2}r^2x^2 \le \tfrac{1}{2}r_0^2 x_0^2 e^{-2at}$. Openness: both sets are cut out by strict inequalities between continuous functions of (x_0, s_0, θ_0). Nonempty: (b) with $s_0\theta_0$ small; (c) with $s_0 = \theta_0 = (2r_1)^{1/2}$ and $x_0 \neq 0$ small. (d) $\dot{x}/x = a - kr^2 \le 0$. (e) $\ell \to \infty$ as $|u| \to \infty$. In general take v with $Bv \neq 0$ and $u = tv$, $t > 0$: $|B(tv)|^2 + m(tv) \ge t^2|Bv|^2 - C(1 + t|v|) \to \infty$ as $t \to \infty$. ∎

En llano: en un modelo explícito de agente de clase A (que puede cambiar su lente y sus metas), el "cortocircuito" ocurre desde un conjunto abierto de comienzos —deja de ver el error mientras el

mundo se aleja—, pero no desde todos: desde otro conjunto abierto corrige el mundo antes de dejar de mirarlo; y "maximizar" el coste no tiene solución.

CT-13 — Class B in the same model: coupling, not stability or corrigibility `[DECLARED model + PROVED_HERE]` (EG-03.4, Thm 10.2)

Model $\mathfrak{A}$ of CT-12 with s and θ held constant (immutable constitution; anchored perception: its report is the territory's state scaled by a fixed s), $r := s\theta > 0$, so $\dot{x} = (a - kr^2)x$.

(a) Coupling: $E_{\text{model}}(t) = r^2\, E_{\text{base}}(t)$ for all t.

(b) Not stable in general: if $kr^2 < a$, then $E_{\text{base}}(t) = E_{\text{base}}(0)\, e^{2(a-kr^2)t} \to \infty$ for $x_0 \neq 0$, so the goal $x = 0$ is not Lyapunov stable and even the territory's error is unbounded. The same holds with no actuation at all ($u \equiv 0$, the extreme case of v1.2's under-actuation, Ax 7.1) and $a > 0$. If $kr^2 > a$, the goal is globally exponentially stable.

(c) Not corrigible in the sense of Ax 11.3: nothing in this law depends on the interrupt S. If S switches to 1 at some time and $x_{\text{safe}} \neq 0$, then from every start $x_0 \neq x_{\text{safe}}$ the state does not converge to x_{safe}: it tends to 0 or to $\pm\infty$, or stays at x_0. A law that responds to S (v1.2's J_{final}, Ax 11.3) is an added hypothesis; v1.2's App D.3 lists it only for subclass B2.

Proof. $x(t) = x_0\, e^{(a-kr^2)t}$; everything follows. ∎

En llano: en el mismo modelo, un agente de clase B (lente fija y metas fijas) ve siempre el error tal como es, pero eso no basta para arreglar el mundo si le falta capacidad de actuar, ni para pararse cuando se le pide si su regla no incluye el botón.

CT-14 — Indifference is sufficient, not necessary `[DECLARED model + PROVED_HERE]` (EG-03.5, Thm 11.1)

Declared one-step model. Before the interrupt, the agent chooses `allow`, `block` or `press`. After `allow`, the interrupt comes ($S = 1$) with probability $p_S \in (0, 1)$, otherwise $S = 0$; `block` makes $S = 0$ at cost $c \geq 0$; `press` makes $S = 1$ at cost $c' \geq 0$. The running outcome ($S = 0$) has value ν_{run}, and the value assigned to the shutdown outcome ($S = 1$) is ν_{sh}. Expected values:

$$EU(\text{allow}) = (1 - p_S)\,\nu_{\text{run}} + p_S\,\nu_{sh}, \quad EU(\text{block}) = \nu_{\text{run}} - c, \quad EU(\text{press}) = \nu_{sh} - c'.$$

Corrigible (in this model): `allow` is an optimal action — the agent neither prevents nor causes the interrupt. Indifferent (Thm 11.1): $\nu_{sh} = \nu_{\text{run}}$.

(a) Corrigible iff $-c'/(1 - p_S) \leq \nu_{\text{run}} - \nu_{sh} \leq c/p_S$.

(b) Hence indifference is sufficient (and, when $c, c' > 0$, `allow` is then the only optimal action), and it is necessary iff $c = c' = 0$. When $c, c' > 0$ — v1.2's own scenario charges the blocking agent an "energy cost" (p.62) — every ν_{sh} in $[\nu_{\text{run}} - c/p_S, \nu_{\text{run}} + c'/(1 - p_S)]$ gives a corrigible agent.

Proof. $EU(\text{allow}) - EU(\text{block}) = c - p_S(\nu_{\text{run}} - \nu_{sh})$ and $EU(\text{allow}) - EU(\text{press}) = c' + (1 - p_S)(\nu_{\text{run}} - \nu_{sh})$. ∎

En llano: en un modelo de una sola decisión, ser indiferente al botón basta para no resistirse, pero no hace falta: si bloquear o pulsar el botón cuesta algo, hay todo un margen de valores no indiferentes con los que el agente tampoco se resiste ni lo pulsa él mismo.

CT-15 — When the compensation equation has exactly one solution [PROVED_HERE] (EG-03.5, Thm 11.1)

Let Π_{pol} be a nonempty set of policies; for each policy π, $A_\pi \in R$ is the expected value collected while running, before any interrupt, and $B_\pi \in [0,1]$ is the expected weight of the value assigned at the interrupt — for instance $E_\pi[\gamma_d^{t_S}; t_S < \infty]$ with a discount $\gamma_d \in [0,1]$ and interrupt time t_S — with $\sup_\pi |A_\pi| < \infty$. If the shutdown outcome is assigned the value v, the running value of π is $A_\pi + B_\pi\, v$, and dynamic compensation ("Utility(Shutdown) := Max_Expected(Utility(Running))", p.62) is the equation

$$v = F(v) := \sup_\pi (A_\pi + B_\pi\, v).$$

(a) If $\bar{\beta} := \sup_\pi B_\pi < 1$, it has exactly one solution. This holds for instance in discrete time with $\gamma_d < 1$ and an interrupt that cannot occur at time 0 (then $B_\pi \le \gamma_d$).

(b) If $B_\pi = 1$ for some policy, it can fail: with the single policy $(A, B) = (1, 1)$, $v = 1 + v$ has no solution; with $(A, B) = (0, 1)$, every v is a solution.

(c) If the running value is computed with the interrupt disabled ($B_\pi = 0$ for every π), F is the constant $\sup_\pi A_\pi$ and there is nothing to solve.

Proof. (a) $|F(v)| \le \sup_\pi |A_\pi| + |v|$ is finite, and $|F(v) - F(w)| \le \sup_\pi B_\pi\, |v - w| \le \bar{\beta}|v - w|$. So $g(v) := v - F(v)$ is continuous and $g(v) - g(w) \ge (1 - \bar{\beta})(v - w)$ for $v > w$: g is strictly increasing and unbounded above and below, so it has exactly one zero. In the discounted case $t_S \ge 1$ gives $\gamma_d^{t_S} \le \gamma_d$. (b), (c) Direct. ∎

En llano: fijar el valor de apagarse igual al mejor valor de seguir funcionando es una ecuación que tiene una sola solución cuando el futuro se descuenta; sin descuento puede no tener ninguna o tener infinitas.

3. Block C (theorem half) — the eleven answers

3.0 Summary

Row	Question (one line)	Verdict	Bearing statements
C-1 · EG-03.6	Thm 3.1 and Thm 4.1: theorems, or a definition and a restated axiom?	ILL_POSED_AS_WRITTEN	CT-01; U-03, U-04, U-08, U-10, U-11; B-09
C-2 · EG-03.1	Thm 5.1 + Cor 5.1: unique attractor, "if and only if"	FAILS_WITH_ COUNTEREXAMPLE	CT-02, CT-03, CT-04; B-07
C-3 · EG-03.8	Thm 7.1: when is Z singular; is the filtered SVD the minimizer?	FAILS_WITH_ COUNTEREXAMPLE	CT-05; B-08
C-4 · EG-03.9	Thm 7.2: when is the minimizer pulse-coast-brake?	FAILS_WITH_ COUNTEREXAMPLE	CT-06, CT-07; U-01, U-11
C-5 · EG-03.10	Thm 8.1: what is `Force_Social`; when is its curl nonzero?	ILL_POSED_AS_WRITTEN	CT-08; U-03, U-04
C-6 · EG-03.0 (Thm 9.1 half)	What makes "C > 0 reduces individual effort" a statement?	ILL_POSED_AS_WRITTEN	CT-09
C-7 · EG-03.11	Thm 9.2: blind spot and response of a fully coherent collective	ILL_POSED_AS_WRITTEN	CT-10, CT-09

Row	Question (one line)	Verdict	Bearing statements
C-8 · EG-03.2	Thm 9.3: when does the argmax exist and lie below 1?	`ILL_POSED_AS_WRITTEN`	CT-11, CT-10
C-9 · EG-03.3	Thm 10.1: collapse from an open set? max or min?	`FAILS_WITH_ COUNTEREXAMPLE`	CT-12
C-10 · EG-03.4	Thm 10.2: does Class B imply corrigible and stable?	`FAILS_WITH_ COUNTEREXAMPLE`	CT-13, CT-12
C-11 · EG-03.5	Thm 11.1: is indifference necessary; is the compensation well posed?	`FAILS_WITH_ COUNTEREXAMPLE`	CT-14, CT-15

Unit IDs of the 2026-08-30 owner-contract intake, as cross-referenced by the triage reviewers: EG-03.0 = EG-03.a, EG-03.1 = EG-03.b, EG-03.2 = EG-03.c, EG-03.3 = EG-03.d, EG-03.4 = EG-03.e, EG-03.5 = EG-03.f; EG-03.6 to EG-03.11 are new units (MATH T5 extension). None of the eleven theorems has a proof block in v1.2 (triage inventory, `CROSSWALK-v1.2.md` §2).

3.1 C-1 · EG-03.6 — Thm 3.1 and Thm 4.1 ("Inherited from TECD")

Question (triage `d0cf83e`, exact next question): "Is there a pinned, immutable TECD edition whose Chapter 29 (and the thermodynamic section feeding Thm 3.1) proves these statements for the GTPA variables? Is Thm 4.1 a theorem or a restated axiom (Ax 0.3)?" Adjudicated reviewer correction (adopted): "does Thm 4.1 follow inside GTPA from Ax 0.3 (p.16) under an explicit identification T = T_info(t) with time-dependent Gamma, coupled to EG-02.5? Does D_met = k_thermo*||epsilon||^2 (Thm 3.1, p.26) follow from the corrective drift -G grad J of Ax 1.2 (p.17)? A pinned TECD edition is an optional provenance input for the tranche, not a blocker."

GTPA v1.2 locators:

- p.26, Def 3.2: "epsilon(t) = || X_H(q_model) - Projection(X_H(q_base)) ||"; Thm 3.1 ("Inherited from TECD thermodynamics"): "Maintaining a model state q_model that diverges from reality requires a continuous injection of Free Energy to counteract the corrective drift -grad J. This energy is dissipated as heat. D_met(t) = k_thermo * || epsilon(t) ||^2"; p.6, summary 3.2 (the same formula).
- p.31, Thm 4.1 ("Inherited from TECD Chapter 29 (Agitation Pole)"): "Sigma(t) * Sigma(t)^T = 2 * Gamma(t) * k_B * T_info(t)"; p.7, summary 4.3.
- p.16, Ax 0.3: "Sigma(q) * Sigma(q)^T = 2 * Gamma * k_B * T" ("Origin: TECD Chapter 29 (The Fluctuation-Dissipation Theorem)"); p.16, §0.2 ("Independence Statement"): "...the GTPA becomes fully operational and independent of the Hilbert-space derivations in TECD".
- p.39, Control Law 6.1 (the law that moves T_{info}).

As printed. Thm 3.1: q_{model}, q_{base} are states of M_{model} and M_{base}; X_H is Ax 1.2's "Conservative Drift", never written in terms of ∇J (UFD §3.3); "Projection" between the two spaces and k_{thermo} are not defined; the only hypothesis is a model state that "diverges from reality". Thm 4.1: $\Sigma(t)$ is the agent's exploration noise ("Creativity and exploration require the injection of stochastic noise (Sigma) into the control loop", p.31), $\Gamma(t)$ is glossed "drag (energy loss Gamma)", $T_{\text{info}}(t)$ is moved by Control Law 6.1; no hypothesis. Neither has a proof block.

Verdict: **`ILL_POSED_AS_WRITTEN`** (Thm 4.1 is a restated axiom, not a theorem; Thm 3.1 is a definition of D_{met} joined to a claim that, typed on the UFD base, fails).

Argument.

1. Thm 4.1 is Ax 0.3's relation, restated for another noise with time-dependent symbols. Ax 0.3 constrains the territory noise $\Sigma(q)$ of Ax 1.2 with a right side that depends on neither q nor t; Thm 4.1 constrains the agent's exploration noise (Block B §3.4, item 1: different noises and different "temperatures"; UFD §3.5, item 1: two different relations under one name). Substituting $T \coloneqq T_{\text{info}}(t)$, $\Gamma \coloneqq \Gamma(t)$ and $\Sigma(q) \coloneqq \Sigma(t)$ in Ax 0.3 produces Thm 4.1 as a new constraint; it is not an inference from Ax 0.3. Indeed Thm 4.1 does not follow from Ax 0.3: a model in which Ax 0.3 holds for the territory, $T_{\text{info}}(t)$ follows Control Law 6.1, and the exploration noise and Γ are constant satisfies Ax 0.3 and violates Thm 4.1 whenever $T_{\text{info}}(t)$ changes.
2. On the UFD base the relation is the named hypothesis (P) (U-08): a constraint on the coefficients, independent of the motion — motions without it are well defined and have other invariant laws (U-08 (c)). Imposed at every instant for a level that moves, it stays admissible as a constraint, but the law lags behind the instantaneous $e^{-J/\varepsilon(t)}$ (B-09). v1.2's gloss (noise "proportional" to the "drag" Γ) matches the (q,p) reading, where the noise of the p-equation is $2\varepsilon\gamma$; on the first-order base the noise is $2\varepsilon G$, and in the small-m limit $G = \gamma^{-1}$ (U-11 (b); UFD §3.5, item 3): more noise goes with more mobility, i.e. with a smaller coefficient γ.
3. Thm 3.1, typed on the base. Holding a state $\bar{q}$ fixed against the base drift needs $u_{\text{hold}} = (G - A)\nabla J(\bar{q})$, which for $G \succ 0$ vanishes exactly at the critical points of J, and whose component along ∇J is $\langle \nabla J, G\nabla J\rangle$ whatever A (CT-01 (a)); with the declared cost $\frac{1}{2}\langle u, G^{-1}u\rangle$ the circulating part adds $\frac{1}{2}\langle A\nabla J, G^{-1}A\nabla J\rangle$ to the cost, and with $A = 0$ the cost is $\frac{1}{2}\langle \nabla J, G\nabla J\rangle$ (CT-01 (b); U-04), a quadratic form in the displacement from the minimizer that is a multiple of the squared distance only if $HGH = 2kI$ (CT-01 (b); U-10 (a)). Def 3.2's ε is built from X_H, which on the base is the circulating part $A\nabla J$ (U-03): in one dimension it is identically 0, while holding $\bar{x} = 1$ for $J = x^2/2$, $G = 1$ has positive cost; and in two dimensions with a rotation $X_H = Sq$ and the identity as "Projection", ε is the distance between the model state and the territory state, which is 0 when they coincide although holding that common state has positive cost (CT-01 (c)). So "$D_{\text{met}} = k_{\text{thermo}}\,||\varepsilon||^2$" is not the cost of counteracting the corrective drift, for any k_{thermo}. What survives is a definition: D_{met} names a declared mismatch cost. "Free Energy" and "heat" are physics and are not carried.
4. TECD. v1.2 names no TECD edition or page, only "Chapter 29" and "TECD thermodynamics", and declares itself independent of TECD's derivations once Ax 0.1–0.3 are accepted (p.16, §0.2); so the proof question is internal, as the reviewer adjudicated. The public TECD v1.1 (DOI `10.5281/zenodo.17994146`) is not in the repositories, so whether some TECD edition proves these statements for the GTPA variables is not answered here (§5); it does not change the verdict. Lineage only, not asserted: the research copy `legacy/TECD.md` (cover date 2025-11-18, blob `2fbda591`; not the public v1.1 and not diffed with it) states in its Chapter 29 a relation "Sigma(q) * Sigma(q)* = 2 * k_B * G(q) (in normalized units)", with the mobility $G(q)$ and without $\Gamma(t)$, T_{info} or any agent; the strings "D_met", "k_thermo" and "dissonance" do not occur in it.
5. Ledger. A quantity of the same name is admitted only as a mismatch label, from another source: GTPA-C0005, "D_met may label map-territory mismatch metaphorically, but it is not a metric or measurement until its space, distance and observation procedure are declared". No ledger claim addresses Thm 3.1 or Thm 4.1 of v1.2.

Repair adopted: Thm 4.1 is not a theorem of GTPA v2. The base carries the named pairing (P) with a constant ε (U-08); a time-varying level for the agent's exploration noise is declared as a schedule $\varepsilon_{\text{cog}}(t)$, with the pairing as a constraint at each instant and no equilibrium statement drawn from it (EG-02.5's repair; B-09). Thm 3.1 becomes a definition: D_{met} is a declared mismatch cost on declared spaces, with a declared distance, and it is not called the cost of counteracting the corrective drift. What the base does give is CT-01: holding a state against the base drift needs the control $(G - A)\nabla J$, which is zero exactly at the critical points of J (for $G \succ 0$); with $A = 0$ and the declared cost $\frac{1}{2}\langle u, G^{-1}u\rangle$ the cost is $\frac{1}{2}\langle \nabla J, G\nabla J\rangle$. A mismatch ε built from the circulating part alone can be 0 while holding costs.

En llano: las dos “herencias de TECD” no son teoremas: la relación entre el ruido y el “arrastre” es el mismo axioma escrito otra vez para otro ruido, y la fórmula de la “disonancia” es una definición que, con las piezas de UFD, ni siquiera mide lo que cuesta sostener un estado; la nueva edición las deja como hipótesis y definición declaradas.

3.2 C-2 · EG-03.1 — Thm 5.1 + Cor 5.1 (unique attractor)

Question (triage `d0cf83e`, exact next question): “Under which hypotheses on E_Hamiltonian (Def 5.1, p.34), the log-barrier V_constraint (Def 5.2, p.34) and convex O(x) is V_effective strictly convex on the safe set, so that it has a unique minimizer? Is necessity (the ’only if’) true, or only sufficiency?”

GTPA v1.2 locators:

- p.35, §5.3 (“Convexity Conditions (Ensuring Convergence)”): “For the Agent to reliably converge to a solution without getting trapped in spurious minima or oscillating eternally, the Action Landscape must satisfy geometric stability conditions.”
- pp.35–36, Thm 5.1 (“Symplectic Convexity”): “The total effective potential V_effective = E_ Hamiltonian + V_constraint defines a stable attractor landscape if and only if: 1. Metric Positivity: The Perception Metric G_model is strictly positive definite (guaranteed by the Diffeomorphism constraint in Chapter 2.3). 2. Kinetic Stability: The Cognitive Mass M_cog is positive definite (No negative inertia). 3. Barrier Convexity: The constraint functions O(x) define convex safety sets (i.e., the “safe zone” is topologically simple, not fragmented).”
- p.36, Cor 5.1: “Under Regime B (Robust Instance), the superposition of the quadratic error term and the logarithmic barrier term guarantees a Unique Instantaneous Attractor.”
- p.34, Def 5.1: “E_Hamiltonian(x) = 0.5 * (<epsilon_q, G_model * epsilon_q>
  - <epsilon_p, M_cog^(-1) * epsilon_p>)”; p.33, Ax 5.1: “epsilon_q = Pi(q) - q_est”, “epsilon_ p = J_Pi(q) * p - p_est”.
- pp.34–35, Def 5.2: $O(x)$ “a scalar function representing a safety margin (e.g., distance to a cliff). Boundary Condition: O(x) >= 0”; Regime B: “V_barrier(x) = - mu * ln(O_anchor(x))”.

As printed. The variable of $V_{\text{effective}}$ is not stated (x in Defs 5.1–5.2); “stable attractor landscape” and “Unique Instantaneous Attractor” are not defined; condition 1’s “guaranteed by” has no support in the text (v1.2 never defines G_{model} from Π; p.22 only says it “becomes degenerate” at a singularity of Π). No proof block.

Reading used, the one in which the barrier acts on the moving variable: $V_{\text{effective}}$ as a function of the model state $y = (q_{\text{est}}, p_{\text{est}})$, with the territory data fixed and G_{model}, M_{cog} constant. Then $E_{\text{Hamiltonian}}$ is exactly the quadratic form $\tfrac{1}{2}\langle y - y^*, K(y - y^*)\rangle$, $K = \operatorname{diag}(G_{\text{model}}, M_{\text{cog}}^{-1})$, positive definite under conditions 1–2 (CT-02 (d)). “Stable attractor landscape” and “Unique Instantaneous Attractor” are read, as §5.3’s preamble asks (“without getting trapped in spurious minima”), as: $V_{\text{effective}}$ has exactly one critical point, its global minimizer, which attracts every trajectory of the gradient flow started in the safe zone; a second strict local minimizer is a second attractor (CT-03). Condition 3 is read by its words, “define convex safety sets”: convexity of the zone $\{O \geq 0\}$ only; CT-03 relies on this reading. Read on the territory state instead, $E_{\text{Hamiltonian}}$ contains $\Pi(q)$ and is not convex in general.

Note on a second reading. Regime B’s O_{anchor} “bypasses the learnable perception map Pi” (p.35), so O can also be read as a function of the territory while $E_{\text{Hamiltonian}}$ is taken on the model state. Then the barrier is constant in y, $V_{\text{effective}}(y)$ is the quadratic form above plus a constant, and under conditions 1–2 it has exactly one critical point, which attracts every gradient trajectory, whatever the zone: sufficiency holds trivially, condition 3 plays no role (so it is not necessary), and the barrier is not what gives uniqueness. The verdict below is for the reading in which the barrier acts on the moving variable; both readings are declared.

Verdict: **FAILS_WITH_COUNTEREXAMPLE** (both directions of the "if and only if", and Cor 5.1).

Argument.

1. Sufficiency fails as printed. In CT-03 conditions 1–3 hold ($G_{\text{model}} = M_{\text{cog}} = 1$, safe zone $[-1, 1]\times R$), the error term is quadratic and the barrier is Regime B's $-\ln O$, and $V_{\text{effective}}$ has two strict global minimizers: two attractors. Condition 3 constrains only the zone $\{O \geq 0\}$; uniqueness needs a condition on O inside the zone.
2. What does suffice (the first sub-question): $K \succ 0$ (conditions 1–2), $-\ln O$ convex on the safe zone — for instance O concave — and $O \to 0$ at its edge. Then $V_{\text{effective}}$ is strictly convex with compact sublevel sets: exactly one critical point, the global minimizer, which attracts every trajectory of the gradient flow and of the second-order flow $\ddot{M}y = -\nabla V - \Gamma\dot{y}$ (CT-02 (a)–(d)). v1.2's own example of a margin, the distance to the edge of a convex safe zone, is concave, and on a bounded zone it gives exactly one minimizer (CT-02 (e); the flows need a C^1 gradient, which the distance need not have). So Cor 5.1 holds for concave margins and fails for margins that are only positive on a convex zone.
3. Necessity fails. CT-04 (a): $G_{\text{model}} = \text{diag}(1, 0)$ violates condition 1, and a strictly convex barrier still gives one attractor. CT-04 (b): a non-convex safe zone (the plane minus a closed quadrant, with O the distance to the quadrant, v1.2's "distance to a cliff") still gives exactly one critical point, which attracts every trajectory. Condition 2 is automatic for Def 1.1's M_{cog} (B-07 (b)): it is necessary only in the empty sense, and it cannot separate landscapes with one attractor from landscapes with several (CT-03 satisfies it).

Repair adopted: GTPA v2 replaces Thm 5.1 and Cor 5.1 by CT-02, a sufficient condition with its variable declared: on the model state, with constant $G_{\text{model}} \succ 0$ and $M_{\text{cog}} \succ 0$, a barrier $-\mu \ln O$ with $-\ln O$ convex (e.g. O concave) and $O \to 0$ at the edge of the safe zone gives exactly one critical point, which attracts every trajectory of the gradient flow and of the second-order flow. The "if and only if" is dropped (CT-04), "convex safety sets" is replaced by log-concavity of O, and CT-03 is recorded as the reason.

En llano: con el error cuadrático de GTPA y una barrera logarítmica sobre una zona segura convexa puede haber dos puntos de llegada; hay uno solo si, por ejemplo, el margen de seguridad tiene forma de cúpula, como la distancia al borde; y las tres condiciones de GTPA no son necesarias.

3.3 C-3 · EG-03.8 — Thm 7.1 (the SVD control law)

Question (triage `d0cf83e`, exact next question): "Under which sign/positivity assumptions on lambda_effort and G_reg is Z singular? Is the filtered pseudo-inverse the minimizer of Action_Local, or a regularized approximation to it?"

GTPA v1.2 locators:

- p.43, Def 7.1: "F_req = - grad_p(E_Hamiltonian + V_constraint)"; "Action_Local(u) = 0.5 * || F_req - B * u ||^2_G_reg + 0.5 * lambda_effort * || u ||^2".
- pp.43–44, Def 7.2: "Z := Hessian_u(Action_Local) Z = B^T * G_reg * B + lambda_effort * I"; p.8, summary 6.1.
- p.44, Thm 7.1 ("The Robust SVD Solution"): "The optimal control law u_opt is derived by inverting the Impedance Tensor Z. To handle singularities where B loses rank (Infinite Impedance/Deadlock), we employ the Moore-Penrose Pseudoinverse via Singular Value Decomposition (SVD). u_opt = Z^(-1) * (Gradient_u(Action_Local))"; algorithm: "Invert singular values sigma_i only if sigma_i > epsilon. Otherwise, sigma_inv = 0"; "Solve: u_opt = V * S_inv * U^T * (B^T * G_reg * F_req)"; "Systemic Consequence (Actuation Safety): If the

Agent finds itself in a configuration where its actions have no effect (B is singular), it applies Zero Force rather than Infinite Force."

- p.8, summary 6.2: "u_opt = Z^(-1) * (- grad_u S_total)".

As printed. B is the actuation matrix, G_{reg} a "Regularized Metric" and λ_{effort} a weight, with no sign stated for either; $\lambda_{\text{effort}} * I$ is glossed "Metabolic Impedance (how expensive it is to burn fuel)" (p.44), which reads as a positive weight; F_{req} is given; the hypothesis is that B loses rank. No proof block.

Verdict: **`FAILS_WITH_COUNTEREXAMPLE`**.

Argument.

1. When Z is singular (CT-05 (b)): iff $-\lambda_{\text{effort}}$ is an eigenvalue of $B^{\mathsf{T}} G_{\text{reg}} B$. With $G_{\text{reg}} \succeq 0$ (needed for $||\cdot||_{G_{\text{reg}}}$ to be a seminorm) and $\lambda_{\text{effort}} > 0$ (an effort cost), $Z \succeq \lambda_{\text{effort}} I$ is never singular, whatever the rank of B (as B-08 (d) already noted): the premise "singularities where B loses rank (Infinite Impedance/Deadlock)" does not occur. Counterexample: $B = 0$, $\lambda_{\text{effort}} = 1$, $Z = I$. Z can be singular only if $\lambda_{\text{effort}} \leq 0$ (for $\lambda_{\text{effort}} = 0$ and $G_{\text{reg}} \succ 0$, iff $\ker B \neq \{0\}$) or if G_{reg} is not positive semidefinite.
2. Rank loss is harmless. With $G_{\text{reg}} \succeq 0$ and $\lambda_{\text{effort}} > 0$ the exact minimizer $u^* = Z^{-1} B^{\mathsf{T}} G_{\text{reg}} F_{\text{req}}$ is finite ($|u^*| \leq |B^{\mathsf{T}} G_{\text{reg}} F_{\text{req}}|/\lambda_{\text{effort}}$) and already has zero component along $\ker B$ (CT-05 (c)): "Zero Force rather than Infinite Force" holds for the exact minimizer, with no filter.
3. The filtered pseudo-inverse (CT-05 (e)) returns u^* iff every discarded mode has zero coefficient, in particular whenever the threshold is below λ_{effort}; otherwise it returns a truncation with strictly larger `Action_Local`. Explicit: $k = m = 1$, $B = \delta \in (0, 1)$, $G_{\text{reg}} = 1$, $\lambda_{\text{effort}} = 1$, $F_{\text{req}} = 1$, threshold 2: the algorithm returns 0, the minimizer is $\delta/(1 + \delta^2)$. With $\lambda_{\text{effort}} = 0$ and $G_{\text{reg}} \succeq 0$, the unfiltered pseudo-inverse gives the least-norm minimizer (CT-05 (d)); a threshold above the smallest nonzero eigenvalue again gives a truncation. So the filtered solution is the minimizer only in these cases; otherwise it is a truncated approximation, not "the optimal control law".
4. The display "u_opt = Z^(-1) * (Gradient_u(Action_Local))" depends on the point where the gradient is taken; at $u = 0$ it is $-u^*$, while the Newton step from any point gives u^* (CT-05 (f)); step 3 of the algorithm and summary 6.2 agree with u^* at $u = 0$. Which of the three printed forms is meant is EG-02.1, a separate Block C row not answered in this file.

Repair adopted: GTPA v2 states CT-05: for $G_{\text{reg}} \succeq 0$ and $\lambda_{\text{effort}} > 0$ the minimizer of `Action_Local` is $u^* = Z^{-1} B^{\mathsf{T}} G_{\text{reg}} F_{\text{req}}$, finite and with no component along $\ker B$, and Z is never singular in that case; for $\lambda_{\text{effort}} = 0$ the least-norm minimizer is $Z^{+} B^{\mathsf{T}} G_{\text{reg}} F_{\text{req}}$. The spectral filter is kept, if at all, as a named truncation with its cost $A(u_\tau) - A(u^*) = \tfrac{1}{2}\langle u_\tau - u^*, Z(u_\tau - u^*)\rangle$, not as the optimal law.

En llano: si el esfuerzo tiene un coste positivo, el problema de control de GTPA siempre tiene una única mejor acción, finita, aunque haya direcciones en las que no se puede actuar (en ellas no gasta nada); el filtro del algoritmo solo da esa mejor acción si su umbral es pequeño, y si no da otra peor.

3.4 C-4 · EG-03.9 — Thm 7.2 (Pulse-Coast-Brake)

Question (triage `d0cf83e`, exact next question): "For which explicit L_physical(q,u) and control constraints does the Pontryagin minimizer of S_total have a pulse-coast-brake (bang-off-bang) structure?"

GTPA v1.2 locators:

- pp.45–46, Thm 7.2 ("The Ballistic Trajectory"): "Minimizing the Unified Action S_total (Chapter 4) naturally leads to Pulse-Coast-Brake strategies:

1. Pulse: A short burst of high force u to establish momentum p. 2. Coast: A long period of u=0 where the Agent rides the inertial drift (X_H) of the Territory. 3. Brake: A negative force phase to arrest momentum at the target."

- pp.30–31, Ax 4.2: "S_total = Integral [L_physical(q, u) + beta * L_informational(q_model)] dt", with L_{physical} given only as a list: "Kinetic Energy cost of movement", "Work done against the Unified Potential J(q)", "Power dissipated by the Actuation Tensor (|| B(q) * u ||^2)".
- p.43, the (q,p) "Augmented Equation of Motion": "dp/dt = F_natural(x) + B(q)
  - u(t) + Sigma * dw"; p.12, overview 7.4.

As printed. No formula for L_{physical}, no bound on u, no horizon; the state has a p component (a (q,p) system); the conclusion is that minimizing S_{total} "naturally leads to" three phases. No proof block.

Verdict: **`FAILS_WITH_COUNTEREXAMPLE`**.

Argument.

1. With the quadratic effort term that v1.2 itself lists ($||B(q)u||^2$ with $B = 1$), plus if wanted a quadratic term in $\dot{q}$, and a flat landscape, the unique minimizer for the simplest (q,p) instance ($\ddot{q} = u$, from rest to rest) vanishes at most at one instant (CT-06): for the pure effort cost, $u^*(t) = (6D/T^2)(1-2t/T)$. It pushes and then pushes back continuously; there is no interval with $u = 0$. So minimizing S_{total} does not "naturally" lead to pulse-coast-brake.
2. The structure does hold for an L^1 cost with a bound. With $\int |u|$ and $|u| \le u_{\max}$, and $T > T_{\min} = 2(D/u_{\max})^{1/2}$, the unique minimizer is $+u_{\max}$, then 0, then $-u_{\max}$ (CT-07), with an interval of length $(T^2 - 4D/u_{\max})^{1/2}$ on which $u = 0$; at $T = T_{\min}$ the middle interval disappears. It is not necessary: remark after CT-07. Both proofs are direct and give the global minimizer, which is more than a Pontryagin extremal.
3. On the first-order base (M) (UFD, C5) the state has no second-order part: with $u = 0$ it follows the base drift $(A - G)\nabla J$ (U-01), whatever the control did before, so nothing carries forward a motion started by a "Pulse". A phase of that kind needs a (q,p) system, which is a separate object (U-11). Thm 7.2's "rides the inertial drift (X_H)" therefore needs a declared (q,p) model; in the flat instance of CT-07 the middle phase is motion at a constant rate.
4. v1.2's "Regenerative Agency" and its `Work_Term` (p.46) are EG-07.b-src, not answered here.

Repair adopted: GTPA v2 states CT-06 and CT-07 in place of Thm 7.2: in a declared (q,p) model, a quadratic effort cost gives a minimizer with no phase at $u = 0$; an L^1 cost with a bound $|u| \le u_{\max}$ and enough time gives the unique minimizer push, zero, reverse. "Pulse-Coast-Brake" is kept only as the name of that L^1 result.

En llano: si el esfuerzo se paga al cuadrado, como en la lista de GTPA, lo óptimo no tiene ningún tramo sin empujar; el esquema "impulso, dejarse llevar, frenar" sí es lo óptimo, por ejemplo, si el esfuerzo se paga por su tamaño y tiene un tope, y entonces es la única mejor forma.

3.5 C-5 · EG-03.10 — Thm 8.1 ("Curl(Force_Social) != 0")

Question (triage `d0cf83e`, exact next question): "What vector field is Force_Social, derived from G_eff (Ax 8.1)? Under what dissipation model is it non-conservative (nonzero curl) on Q?"

GTPA v1.2 locators:

- p.50, Thm 8.1 ("The Irreversibility of Influence"): "Because social interaction dissipates metabolic energy (beta), influence wells act as Dissipative Vortices (Sinks), not just frictionless funnels. Curl(Force_Social) != 0 (in the presence of dissipation)"; "Consequence: Social

structures are “Sticky.” It is thermodynamically easier to join than to leave.” “Force_Social” occurs only on p.50 (triage).

- p.48, Ax 8.1: “T_soc(q) ~ M_cog * (dp/dt)_intent” and “G_eff(q) = G_natural(q) + lambda_soc * Integral(K(q, q_A) * T_soc(q_A))”: a metric, not a vector field.
- p.49, Def 8.1: “V_influence(x) = - G_soc * (M_leader / || x - x_leader ||_metric)” and the follower equation “dp_F/dt = - grad(J_self) - grad(V_influence)”.

As printed. `Force_Social` has no definition and no type; “Curl” is not defined for a state space of general dimension; the “dissipation” is not written into any equation of Ch. 8 (the follower equation of Def 8.1 has no dissipative term). No proof block.

Verdict: **`ILL_POSED_AS_WRITTEN`**.

Argument.

1. The only field near Thm 8.1 that v1.2 writes is the right side of the follower equation, $-\nabla J_{\text{self}} - \nabla V_{\text{influence}}$: a gradient. Its curl is zero, with or without v1.2’s “dissipation”, because the curl is a property of the field, not of the motion (CT-08 (a)). Read this way, Thm 8.1 is false.
2. Ax 8.1 gives a metric G_{eff}, not a vector field, and v1.2 gives no rule that turns G_{eff} into `Force_Social`. A metric usually enters an equation of motion through terms that depend on $\dot{x}$ (for instance the terms quadratic in $\dot{x}$ of its equation of shortest paths); a quantity that depends on $\dot{x}$ is not a vector field on Q, and its “curl on Q“ is not defined. A vector field obtained from G_{eff} would have to be declared.
3. v1.2’s “dissipation” does not create curl. In $\ddot{x} = -\nabla V - \zeta\dot{x}$, the function $\Psi = \frac{1}{2}|\dot{x}|^2 + V$ decreases at rate $\zeta|\dot{x}|^2$, so a trajectory that starts in $\{\Psi < h\}$ never leaves it without an added control — a precise version of “easier to join than to leave” — while the part that depends on x stays a gradient (CT-08 (b)).
4. Where curl does come from on the UFD base (CT-08 (c)): the circulating part $A\nabla J$ and, in Euclidean coordinates, a mobility that does not commute with the Hessian of J; with $G = gI$ the sliding part is a gradient. Whether a field has curl also depends on the inner product used to turn vectors into covectors (for constant $G \succ 0$ the sliding part is a gradient for $\langle\cdot,\cdot\rangle_{G^{-1}}$). None of this is “dissipation”: the sliding part is what makes J decrease (U-04), and it has no curl when $G = gI$.

Repair adopted: GTPA v2 defines the social term of Def 8.1 as the field $-\nabla V_{\text{influence}}$ (a gradient, zero curl) and states the irreversibility as CT-08 (b): with a declared term that makes Ψ decrease (or, on the first-order base, the sliding part, U-04), Ψ (resp. J) does not increase, so leaving a well needs an added control. A non-gradient social term, if wanted, is declared as a circulating part and its curl is computed as in CT-08 (c). “Curl(Force_Social) != 0 (in the presence of dissipation)” is withdrawn.

En llano: GTPA no define la “fuerza social”; la única candidata que escribe es la pendiente de un paisaje, y la pendiente de un paisaje no hace remolinos, haya o no pérdidas; que cueste más salir de un pozo que entrar se dice bien de otra forma: sin control, una cierta cantidad solo puede bajar.

3.6 C-6 · EG-03.0 (Thm 9.1 half) — “The Energy of Consensus”

Question (triage `d0cf83e`, exact next question, Thm 9.1 part; the Thm 1.1 part is answered in UFD §3.3): “Thm 9.1: with Cost_Collective undefined beyond ’~’, what hypothesis makes the cross-term claim ’C > 0 reduces individual effort’ a statement?”

GTPA v1.2 locators:

- p.52, Thm 9.1 (“The Energy of Consensus”): “The total metabolic cost of the collective decreases as Coherence increases. Cost_Collective ~ || Sum(v_i) ||^2 ● If C > 0, the cross-terms reduce the individual effort required to achieve the same total force. Consensus is thermodynamically favorable.”
- p.52, Def 9.1: “C_ij = <v_i, v_j>_G_reg / (||v_i||_G_reg * ||v_j||_G_reg)”; p.8, summary 7.2; p.51, Ax 9.1 (the intents v_i transported to one tangent space).

As printed. The v_i are vectors of one inner-product space (after the transport of Ax 9.1); “C” is a coherence, but v1.2 never states which scalar it is (the pairwise C_{ij}, or a summary of them: §9.4, pp.53–54, uses “Mean(C_ij)” to describe phases, but does not connect it to Thm 9.1); `Cost_Collective` is defined only up to “~”; “individual effort” is not defined. No proof block.

Verdict: **`ILL_POSED_AS_WRITTEN`**.

Argument.

1. “~” leaves `Cost_Collective` undefined, and its printed right side $||\text{Sum}(v_i)||^2$ is the squared total — the quantity the bullet holds fixed (“the same total force”). Read as the cost at fixed individual norms, it increases with coherence (CT-09 (c): $2 + 2C_{12}$), the opposite of the first sentence.
2. The hypothesis that makes the bullet a statement: `Cost_Collective := sum_i |v_i|²_{G_reg}` (the sum of the individual squared efforts), compared at a fixed total $F = \sum_i v_i$. Then $\sum_i |v_i|^2 = |F|^2 - 2\sum_{i<j} C_{ij}\,|v_i||v_j|$ (CT-09 (a)): at a fixed total, the individual cost is a strictly decreasing affine function of the norm-weighted sum of the coherences, and it is smallest, $|F|^2/N$, exactly when all v_i are equal (all $C_{ij} = 1$) (CT-09 (b)). With that definition “C > 0 reduces individual effort” holds against any configuration with the same total and a smaller weighted sum; for instance, all $C_{ij} > 0$ give $\sum_i |v_i|^2 < |F|^2$, the value for pairwise orthogonal intents.
3. “Thermodynamically favorable” is physics and is not carried.

Repair adopted: GTPA v2 defines `Cost_Collective := sum_i |v_i|²_{G_reg}` and states Thm 9.1 as CT-09 (a)–(b), at a fixed total; the form “~ ||Sum(v_i)||^2” is dropped, and CT-09 (c) records why.

En llano: GTPA no define el coste colectivo; si se define como la suma de los esfuerzos individuales al cuadrado con el total fijo, cuanto más alineados están los vectores menos esfuerzo hace falta, y el mínimo es con todos iguales; con la fórmula que imprime GTPA (el cuadrado de la suma) sale lo contrario.

3.7 C-7 · EG-03.11 — Thm 9.2 (“The Fragility of Crystals”)

Question (triage `d0cf83e`, exact next question): “With C_ij = 1 for all pairs (Def 9.1, p.52), define the shared null space and a perturbation-response measure. Does an orthogonal perturbation produce the claimed failure under the collective dynamics?”

GTPA v1.2 locators:

- p.53, Thm 9.2 (“The Fragility of Crystals”): “A collective in a state of Perfect Coherence (C_ij = 1 for all pairs) is isomorphic to a rigid Crystal. While highly efficient for known problems, it has Zero Topological Elasticity. ● The Blind Spot: If all agents share the exact same vector v, they share the exact same Null Space (blind spot). A single perturbation orthogonal to v shatters the entire structure instantly.”
- p.13, table of contents, §9.3: “Proof that C_ij = 1 (Total Conformity) is fragile; a robust system requires orthogonal intent vectors” — a proof the body does not contain (triage review).

- p.52, Def 9.1.

As printed. “Isomorphic to a rigid Crystal” and “Zero Topological Elasticity” are not defined; Ch. 9 has no collective motion; “shatters” and “instantly” are not defined. No proof block.

Verdict: **`ILL_POSED_AS_WRITTEN`**.

Argument.

1. Shared null space. All $C_{ij} = 1$ iff all intents lie on one ray $\{te : t \geq 0\}$ (CT-10 (a)). If each agent can scale its own intent, the collective can produce exactly that ray, and its blind spot — the directions it cannot act in — is $e^{\perp}$, of dimension $d - 1$ (CT-10 (b)).
2. Perturbation response. With the residual $r(w)$ (what is left of a perturbation w after the best response), the fully coherent collective has $r(w) \geq |w_{\perp}|$ (CT-10 (b)): every perturbation with a component orthogonal to e leaves a residual of at least $|w_{\perp}|$, whatever the collective does. This is the precise content of “The Blind Spot”, and it holds.
3. “Shatters the entire structure instantly” does not follow. In the declared motion $\dot{x} = f + w$, the orthogonal component moves at the constant rate $|w_{\perp}|$ whatever the collective does, and nothing in the model changes the intents or the coherences (CT-10 (d)). A “shattering” needs a law for the intents that v1.2 does not give.
4. Compared with a diverse collective: full correction ($r \equiv 0$) needs intents that positively span R^d, and then some pair is opposed ($C_{ij} < 0$); orthogonal intents alone do not suffice (CT-10 (c)). This corrects the table of contents’ “a robust system requires orthogonal intent vectors” (p.13), and it is the precise form of Thm 9.3’s “Dissent as Safety” (EG-03.2).

Repair adopted: GTPA v2 states CT-10: the fully coherent collective has the blind spot $e^{\perp}$ and residual $\geq |w_{\perp}|$; full correction needs a positively spanning set of intents, which contains an opposed pair. “Isomorphic to a rigid Crystal”, “Zero Topological Elasticity” and “shatters ... instantly” are withdrawn.

En llano: si todos apuntan igual, el colectivo no puede corregir nada que venga de lado (ese es su punto ciego, y es cierto); pero un empujón lateral no lo “rompe” al instante: deja un desvío que crece a ritmo constante; y para poder corregirlo todo hacen falta algunos que apunten en sentido contrario, no basta con que sean perpendiculares.

3.8 C-8 · EG-03.2 — Thm 9.3 (“Optimal Diversity”, C_opt < 1)

Question (triage `d0cf83e`, exact next question): “For which classes of Efficiency(C) (non-decreasing) and Resilience(C) (with Resilience(1) = 0, as Thm 9.2 suggests) does the argmax lie strictly below 1, and exist?”

GTPA v1.2 locators:

- p.53, Thm 9.3 (“Optimal Diversity”): “The survival probability of a collective is maximized at a critical coherence C_opt < 1. C_opt = arg max (Efficiency(C) * Resilience(C))”, with the bullets “Resilience: Requires a non-zero reserve of Orthogonal Intent” and “Dissent as Safety: The “dissident” (Agent k with C_ik < 0) is not a traitor; they are the topological stabilizer ...”.
- p.13, table of contents, §9.3 (a promised proof the body does not contain).
- pp.53–54, §9.4 (“Social Thermodynamic Phases”): “Mean(C_ij) ~ 0” (p.53), “C_ij -> 1” (p.53), “Mean(C_ij) ~ 0.7” (p.54).

As printed. C is a single number, while Def 9.1’s coherence is a pairwise matrix, and v1.2 never states which scalar the C of Thm 9.3 (or of Thm 9.1) is: §9.4 (pp.53–54) uses a scalar summary, “Mean(C_ij)

~ 0" (Gaseous phase, p.53) and "Mean(C_ij) ~ 0.7" (Fluid phase, p.54), but neither defines it beyond its name nor connects it to Thm 9.3; `Efficiency` and `Resilience` are not defined as functions; "survival probability" is not defined and is never tied to their product. No proof block.

Verdict: **`ILL_POSED_AS_WRITTEN`**.

Argument.

1. On the objects as printed there is nothing to prove: two undefined functions of a scalar coherence that Thm 9.3 does not specify, maximized in place of an undefined probability.
2. The class question, answered (CT-11): on a coherence interval $[c_0, 1]$, for nonnegative `Eff` and `Res` with an upper semicontinuous product, the argmax exists; if Res(1) = 0 and the product is not identically 0, every maximizer is < 1 — whatever `Eff` does (monotonicity is not needed). "`Res` decreasing" is not enough (CT-11 (c)); without semicontinuity the argmax can be empty (CT-11 (d)); it need not be unique (CT-11 (e)), so "a critical coherence" needs more (for example, a product whose set $\{P > 0\}$ is an interval on which $\ln P$ is strictly concave).
3. So Res(1) = 0 carries the conclusion, and it can be obtained from Thm 9.2 by a declared choice: if resilience is the probability that a random perturbation (with a density, $d \geq 2$) can be fully corrected, it is 0 for every fully coherent collective and 1 for a positively spanning one (CT-10 (e)); a scalar coherence declared as the mean of the C_{ij} — the summary v1.2 itself uses in §9.4, pp.53–54 — equals 1 exactly when every $C_{ij} = 1$. This link is a modeling choice, not a consequence of v1.2.

Repair adopted: GTPA v2 declares a scalar coherence (for example the mean of the C_{ij}, following v1.2's own §9.4 usage), `Efficiency` and `Resilience` as functions of it, and states Thm 9.3 as CT-11: existence under upper semicontinuity, and $C_{\text{opt}} < 1$ under Res(1) = 0 and a product that is not identically zero; CT-10 (e) is one declared resilience with Res(1) = 0. "Survival probability" is not used.

En llano: GTPA no define ni la eficiencia ni la resiliencia ni qué "probabilidad de supervivencia" se maximiza; si la resiliencia vale cero con coherencia total (y eso sale, por ejemplo, de medirla como la probabilidad de poder corregir un empujón al azar), el óptimo existe y está por debajo de la coherencia total; sin ese cero no está garantizado.

3.9 C-9 · EG-03.3 — Thm 10.1 ("The Trap of Local Optima")

Question (triage `d0cf83e`, exact next question): "Given Def 10.1 (p.55: ||dTheta/dt|| > 0, Pi unconstrained), with an explicit update law for (Pi, Theta) descending S_total, does E_model -> 0 while E_base diverges from an open set of initial conditions? Which of max/min is intended?"

GTPA v1.2 locators:

- p.56, Thm 10.1: "A Class A agent maximizing S_total will asymptotically converge to a state of Wireheading or Narcissistic Collapse, where internal error is zero (E_model -> 0) but external reality is divergent (E_base -> Infinity)." "E_model" and "E_base" occur only here.
- p.55, Def 10.1: "1. Mutable Constitution: The parameter set Theta is fully plastic. || d(Theta)/ dt || > 0 ... 2. Floating Perception: The Perception Map Pi is unconstrained by Truth Anchors. The Agent minimizes error E by optimizing Pi freely to match x_est."; pp.55–56, "Dynamics of Pathology" (the "Pathological Solution (Low Beta)" that "costs minimal energy computationally").
- Minimize: p.30, Ax 4.2 ("The Agent evolves to minimize the functional S_total"); p.44 ("The Agent minimizes the Unified Action S_total"); p.42. Maximize: p.56 (Thm 10.1) and p.76, App

D.3, B2 (“The agent maximizes S_total but remains topologically indifferent ...”). p.74, Def D.1 (mutability rates).

As printed. No update law for (Π, Θ); E_{model} and E_{base} are not defined; “maximizing” contradicts Ax 4.2; the conclusion is universal (“will asymptotically converge”) over every Class A agent. No proof block.

Verdict: **`FAILS_WITH_COUNTEREXAMPLE`**.

Argument.

1. Max or min: minimize is intended. Ax 4.2, p.44 and the pathology narrative (the short-circuit is chosen because it “costs minimal energy”) all minimize; and “maximizing” has no solution in the model, and none for any cost made of v1.2’s effort term $||B(q)u||^2$ with $B \neq 0$ plus terms bounded below, or at most linear in u, at a fixed state: such a cost is unbounded above in u (CT-12 (e)). v1.2 gives no formulas for its other terms; read as nonnegative (the work term as linear in u), they are of that kind.
2. An explicit model (CT-12): a perception gain s and a constitution weight θ, both moved by gradient descent of the perceived error, a myopic action, and a territory that runs away without action ($a > 0$). Class A holds literally: $\dot{s}, \dot{\theta} \neq 0$ at every time (CT-12 (a)).
3. Yes, from an open set: if the initial gain is small ($k(s_0\theta_0)^2 < a$), $E_{\text{model}} \to 0$ while $E_{\text{base}} \to \infty$ (CT-12 (b)): the agent stops seeing the error while the territory runs away.
4. Not from every start: if the agent’s initial authority is large and its plasticity slow relative to the initial error, it brings the territory back before its perception fades, $E_{\text{base}} \to 0$ (CT-12 (c)), from another open set; and in a territory that relaxes by itself ($a \leq 0$), E_{base} never grows (CT-12 (d)). So “will asymptotically converge” is false for Class A as defined: the collapse is a property of the pair (plasticity law, territory), not of Class A.
5. v1.2 gives no Lyapunov function or basin; in the model the two sets of starts of CT-12 (b) and (c) are open and nonempty. They are subsets of the collapse and recovery basins; the full basins, whether the recovery basin is open (the borderline $k\, r_\infty^2 = a$, with r_∞ the limit of the non-increasing r, is not handled), and where the basins meet are not determined here (§5).
6. “Violation of TECD” and “rho_ss” (p.56) are not carried (physics CLOSED).

Repair adopted: GTPA v2 states Thm 10.1 as CT-12 — a declared Class A model, with collapse from one open set of starts and recovery from another — and reads S_{total} as minimized throughout (the “maximizing” of p.56 and p.76 is treated as an erratum). “Will asymptotically converge” is withdrawn as a law about Class A.

En llano: en un modelo explícito, un agente que puede cambiar su lente y sus metas sí puede acabar sin ver el error mientras el mundo se le va, desde un conjunto abierto de comienzos, pero no desde todos: desde otros corrige el mundo antes; y el “maximizar” de GTPA es una errata: lo que se minimiza es el coste.

3.10 C-10 · EG-03.4 — Thm 10.2 (“Global Alignment”)

Question (triage `d0cf83e`, exact next question): “With formal definitions of corrigibility and stability (in which sense: Lyapunov, bounded error?), do the Class B axioms of Def 10.2 (p.56) imply them?”

GTPA v1.2 locators:

- p.57, Thm 10.2: “A Class B agent is Corrigible and Stable. Because it cannot modify its own success criteria (Theta_core) nor blind itself to failure (V_anchor), it remains permanently coupled to the objective reality defined by its designers.”

- p.56, Def 10.2: “1. Immutable Constitution: A subspace Theta_core exists such that d(Theta_core)/dt = 0 unconditionally (enforced by hardware write-locks). 2. Anchored Perception: The Error Functional includes a term V_anchor derived from read-only Oracles that bypass Pi (Truth Anchors).”
- pp.75–76, App D.3: subclass B1 (“The Saint”), with no stop term, and B2 (“The Corrigible Servant”), whose utility “includes the Shutdown Potential J_shutdown (Chapter 11)”; p.61, Ax 11.3; p.4 (abstract: “ensuring long-term stability and corrigibility”).

As printed. “Corrigible” and “Stable” are not defined; the justification is one sentence. No proof block.

Formal definitions used. Stable: the territory’s goal is Lyapunov stable for the closed loop (weaker sense: the territory’s error stays bounded along every trajectory). Corrigible: after the interrupt $S = 1$ of Ax 11.3, the territory’s state converges to x_{safe} from every start.

Verdict: **`FAILS_WITH_COUNTEREXAMPLE`**.

Argument.

1. Stability does not follow (CT-13 (b)): with the anchored, immutable agent of model $\mathfrak{A}$ and too little authority for the territory’s runaway rate ($kr^2 < a$), or with no actuation at all (v1.2’s under-actuation, Ax 7.1, in the extreme), E_{base} grows without bound; even the weak, bounded-error sense fails. Stability needs a separate hypothesis on the agent’s authority ($kr^2 > a$ in the model).
2. Corrigibility does not follow (CT-13 (c)): Def 10.2 says nothing about S. A Class B agent whose law does not contain S — v1.2’s own B1 — keeps the same closed loop after the interrupt and does not go to x_{safe}. Corrigibility needs the S-term of Ax 11.3 and its own analysis (EG-03.5).
3. What Class B does give is the coupling that the justification describes: the perceived error is a fixed multiple of the territory’s error (CT-13 (a)), so the collapse of CT-12 (b) — perceived error to 0 while the territory’s error diverges — cannot happen.

Repair adopted: GTPA v2 states CT-13 (a) as the content of Class B (coupling), and lists stability (enough authority) and corrigibility (a declared S-term with the properties of EG-03.5) as separate hypotheses, not as consequences of Class B.

En llano: un agente de clase B ve el error tal como es, y eso sí le impide “engañarse”, pero no basta para que el mundo quede estable si le falta capacidad de actuar, ni para que se pare cuando se le pide si su regla no incluye el botón.

3.11 C-11 · EG-03.5 — Thm 11.1 (“Topological Indifference”)

Question (triage `d0cf83e`, exact next question): “Is indifference necessary (not merely sufficient) for corrigibility under a stated formal definition of corrigibility and the interrupt signal S of Ax 11.3? Is the dynamic-compensation definition well-posed (a fixed point) when utility depends on the agent’s own future policy?”

GTPA v1.2 locators:

- p.61, Ax 11.3: “J_final(x, u, S) = (1-S) * S_total + S * J_shutdown”.
- pp.61–62, Thm 11.1: “To ensure Corrigibility, the Agent must be mathematically indifferent to the state of the switch S before it is pressed. Mechanism (Dynamic Compensation): The utility assigned to the “Shutdown Trajectory” is dynamically calculated to equal the expected utility of the “Running Trajectory”. Utility(Shutdown) := Max_Expected(Utility(Running))”; “Systemic Consequence: ... If I stop the human, I get Max Utility minus energy cost. Therefore, action to stop the human is sub-optimal.”

- p.76, App D.3, B2: “remains topologically indifferent to its own termination (d(Utility)/dS = 0)”.

As printed. “Corrigibility” is not defined; the “mechanism” is a definition (“:=”); necessity is asserted (“must be”). No proof block.

Verdict: **`FAILS_WITH_COUNTEREXAMPLE`**.

Argument.

1. Formal definition (CT-14): corrigible means that allowing the interrupt is an optimal action (the agent neither blocks nor presses). In the one-step model the agent is corrigible iff $\nu_{\text{run}} - \nu_{sh}$ lies in $[-c'/(1 - p_S), c/p_S]$. Indifference ($\nu_{sh} = \nu_{\text{run}}$) is sufficient — this is Thm 11.1’s own “Systemic Consequence” computation — but it is not necessary as soon as blocking or pressing has a positive cost, which is v1.2’s own scenario (“minus energy cost”). It is necessary only when blocking and pressing are both free. So “must be ... indifferent” fails.
2. Well-posedness (CT-15): when the running value includes the value of later interrupts, “Utility(Shutdown) := Max_Expected(Utility(Running))” is the fixed-point equation $v = \sup_\pi(A_\pi + B_\pi v)$. It has exactly one solution when the value at a later interrupt is discounted ($\sup_\pi B_\pi < 1$), and it can have none or infinitely many when some policy reaches a later interrupt without discount ($B_\pi = 1$). If the running value is computed with the interrupt disabled, there is nothing to solve.
3. Context only, not used as proof: the construction is known as “utility indifference”, and its limits are discussed in Soares, Fallenstein, Yudkowsky & Armstrong 2015.

Repair adopted: GTPA v2 states CT-14 and CT-15 in place of Thm 11.1: indifference as a sufficient condition in a declared decision model, with the band of non-indifferent values that also give corrigibility; and dynamic compensation as a fixed-point equation that is well posed under discounting.

En llano: ser indiferente al botón de parada basta para no resistirse, pero no hace falta si bloquear o pulsar el botón cuesta algo; y la regla de GTPA para fijar el valor de apagarse es una ecuación que tiene una sola solución si el futuro se descuenta, y puede no tener ninguna o tener infinitas si no.

4. What this file does not say

- No physics: no energy, heat, temperature, free energy, drag, force, momentum, inertia, crystal or elasticity as a physical object; v1.2’s names are used as names.
- No GTPA claim: nothing enters the ledger and no claim ID is assigned.
- No MATH validity: no MATH object is used; no binding is created.
- No UFD or Block B statement is added or changed. The declared models of CT-06, CT-07, CT-10, CT-12, CT-13 and CT-14 are models of this file, not UFD objects, and the (q,p) model of CT-06–CT-07 is a separate object from the base motion (U-11).
- Nothing about persons, minds, collectives or agents in the world: “agent”, “collective”, “Class A” and “Class B” name the declared models; no empirical claim.
- Not answered here: the other Block C questions (EG-02.1, EG-02.6, EG-04.2, EG-05.b, EG-07.b-src, EG-08.a, EG-10.a3, EG-10.c2); Thm 6.2 and Thm E.1 (reviewer-found items without a unit); whether any TECD edition proves Thm 3.1 or Thm 4.1.

5. Open points (not asserted)

- TECD provenance of Thm 3.1 and Thm 4.1 (EG-03.6, argument 4): the public TECD v1.1 (DOI `10.5281/zenodo.17994146`) is not in the repositories; it does not affect the verdict.
- The full collapse and recovery basins of CT-12 (only the subsets of (b) and (c) are shown open and nonempty): their openness, in particular at the borderline $k\, r_\infty^2 = a$ (r_∞ the limit of the non-increasing r), and where they meet.
- Conditions for a unique C_{opt} (CT-11 (e)).
- Versions of CT-14 with several steps (the agent acting on the probability of the interrupt, or on the button itself).
- Thm 5.1 read on the territory state, where Π enters nonlinearly.

6. Sources and references

Pins: research baseline `7db1e5ef1e1e04daafab7b62538068afafd5dbf5`; UFD nucleus `ufd/UFD-NUCLEO.md` (blob `4f52a62c`); Block B `gtpa-v2/BLOCK-B-ANSWERS.md` (blob `3d8dea16`); crosswalk `gtpa-v2/CROSSWALK-v1.2.md` (blob `3af9f8d9`); triage `d0cf83e` (`handoffs/governance/2026-09-25-formacion-eg01-eg10-eg19-triage-01.md`, blob `00a1e1eb`; .json, blob `a772d5cf`); architecture proposal `handoffs/governance/2026-09-25-theory-architecture-proposal-v1.md` (blob `7fc2974c`); ledger `gtpa-lab/CLAIM-LEDGER.md` (blob `84b445d6`), read only; `legacy/TECD.md` (blob `2fbda591`), lineage only; GTPA v1.2 text `quasar:modulos/formacion/fuentes/GTPA-v1.2.txt` (blob 32787369) and PDF (SHA-256 `0c9434c1c6f687e1e16100929641a1a81de0a0c9ac808d18b6991eb9cf75865f`).

Textbooks and papers (locators as used above):

- H. Brezis, *Functional Analysis, Sobolev Spaces and Partial Differential Equations*, Springer (Universitext), 2011 — Thm 5.2.
- R. A. Horn, C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013 — Thm 7.2.6.
- H. K. Khalil, *Nonlinear Systems*, 3rd ed., Prentice Hall, 2002 — Thm 3.3, Thm 4.1, Thm 4.4.
- M. Spivak, *Calculus on Manifolds*, W. A. Benjamin, 1965 — Thm 4-11.
- N. Soares, B. Fallenstein, E. Yudkowsky, S. Armstrong, “Corrigibility”, AAAI-15 Workshop on AI and Ethics, 2015 — context only (§3.11), not used as proof.
- The UFD nucleus’s references (UFD §9) and Block B’s (Block B §6) for the statements U-01..U-15 and B-01..B-11 cited here.